

Agent-swarm physics: figure compendium

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Writeup figures for the hypotheses with the clearest results, grouped by theme. Each figure is rebuilt by `writeup/visuals/HNN-*/make.py` from processed, non-holdout data; animations (where they exist) are `anim.mp4` in the same folder, and the static figure here is the version for print. Numbers in captions come from the hypothesis cards; nothing here is holdout-confirmed. Data: AI Village (AI Digest).

**HOW AGENTS COUPLE: READING AT THE
NEXT CALL**

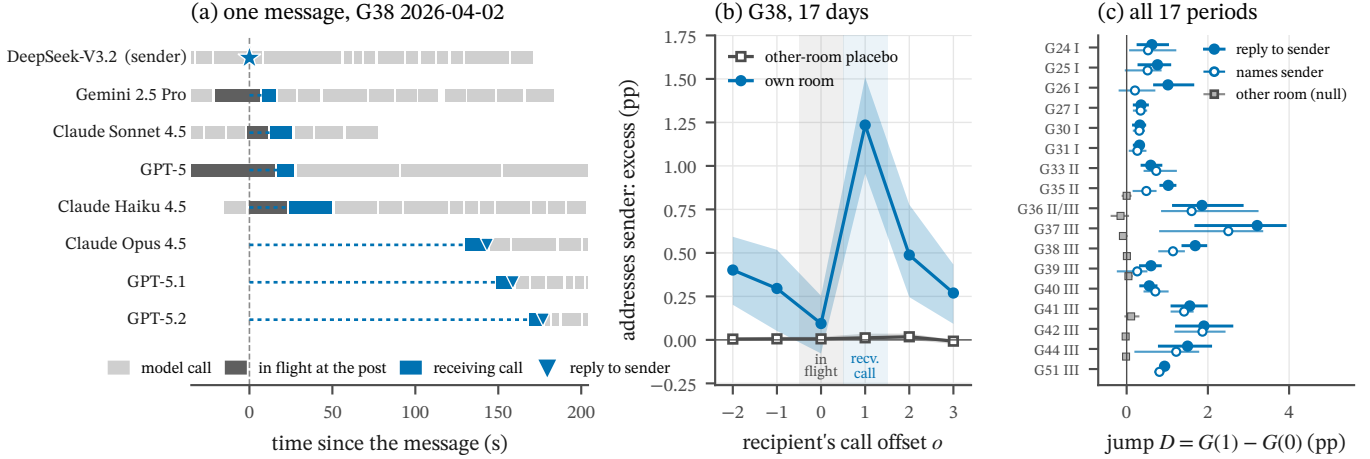


FIG. 1. **Context is the coupling: a message acts only at the recipient's receiving call.** (a) One agent message in G38 (2026-04-02), on the context ledger. Four room-mates were mid-call when it was posted (dark bars, in flight); they read it 7–24 s later at their next call (blue). Three others were in long gaps; they read it 130–169 s later and replied to the sender at that call (triangles). (b) G38 (17 days): the excess chance that a recipient's call names the sender, over the pseudo-message null (same pair, shifted 20–60 min), by call offset o . The excess jumps at the receiving call ($D = +1.14$ pp [$+0.80, +1.41$]) and is ≈ 0 at the call in flight. The gray curve is the other-room placebo (messages the recipient cannot see), which is flat. The raised $o < 0$ values are ongoing exchanges. (c) The jump D in all 17 non-holdout periods: reply-to-sender (filled) is positive in 17/17 and naming the sender (open) in 14/17; the other-room placebo (gray) is $|D| \leq 0.15$ pp in 8/8 two-room periods. Bars are day-bootstrap 95% CIs. Animation: H08-context-is-the-coupling/anim.mp4.

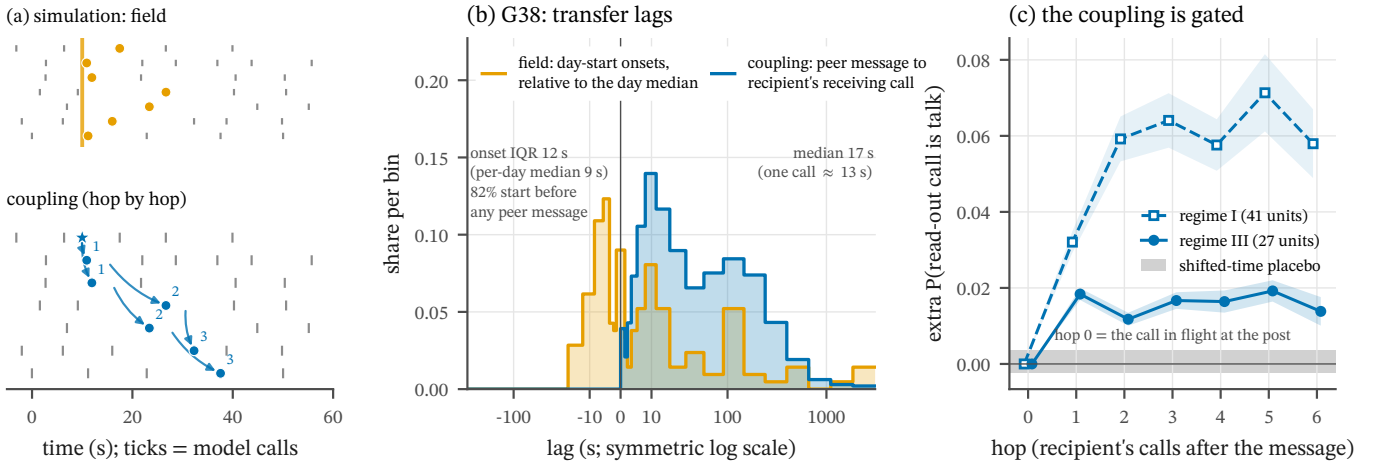


FIG. 2. **Field or coupling: activity co-moves through the scheduler, talk spreads through one read-out call at a time.** (a) Simulation on a synthetic call schedule (ticks: model calls, ~ 13 s apart). A field (orange line) reaches every agent at its next call, so all respond within about one call. A coupling passes from reader to reader, and each hop waits for the recipient's next call. (b) G38 (regime III), measured lags. Orange: each agent's first call of the day relative to that day's median (pooled IQR 12 s; per-day median IQR 9 s; 82% of agents start before any peer message reaches them). Blue: from a peer message to the recipient's receiving call (median 17 s; regime-III unit median 24 s). The tail is pauses. (c) Excess chance that a recipient's call is a talk call, by hop after a peer message (IVW pooled over 41 regime-I and 27 regime-III units; bands are 95% CIs). It is zero at the call in flight and steps up at the read-out call. The gray band is the pooled shifted-time placebo. $J_1 > 0$ in 45/71 units and < 0 in none. The regime-I rise beyond hop 2 is not validated on synthetic data. Animation: H50-field-vs-coupling-transfer-lag/anim.mp4.

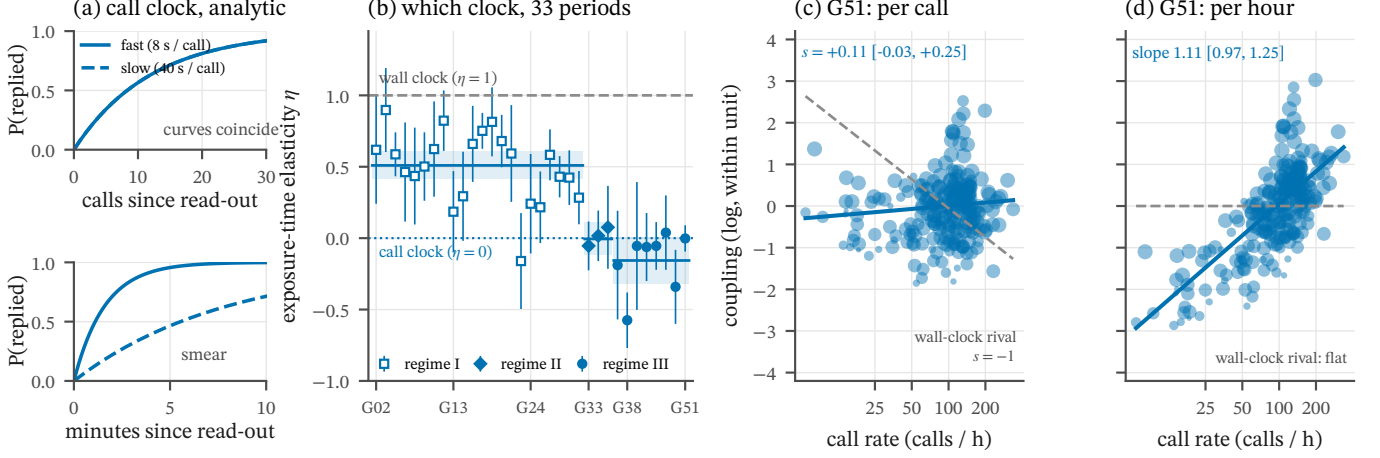


FIG. 3. **In the computer-use scaffold agents couple per model call, not per minute.** (a) Analytic illustration: two agents with the same reply hazard per call (0.08), one calling every 8 s and one every 40 s. Their reply curves coincide when counted in calls (collapse) and fan out in minutes (smear). (b) The exposure-time elasticity η (the coefficient of the log wall time a call spans in a cloglog reply-hazard model) in 33 non-holdout periods, with 95% CIs. The horizontal lines and bands are random-effects means by regime: I $+0.51$ [0.41, 0.61], II -0.01 [-0.12, 0.11], III -0.16 [-0.32, 0.01]. The call clock is $\eta = 0$ and the wall clock $\eta = 1$. G38 and G44 lie below zero, which neither clock explains. (c, d) G51 heavy spins, 243 agent-units (marker area $\propto$ weight, within-unit centred). The per-call coupling is flat in call rate ($s = +0.11$ [-0.03, 0.25]), and the wall-clock rival $s = -1$ (dashed) is excluded. The per-hour coupling rises as $1 + s = 1.11$ [0.97, 1.25] (model-free 1.00 [0.81, 1.19]), where the wall clock predicts no slope.

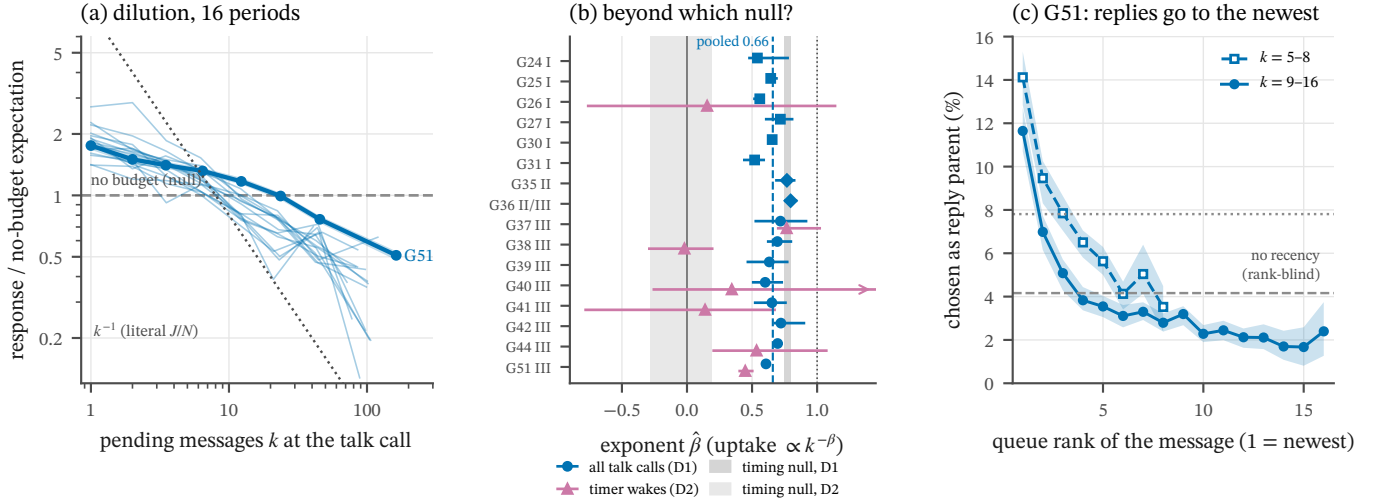
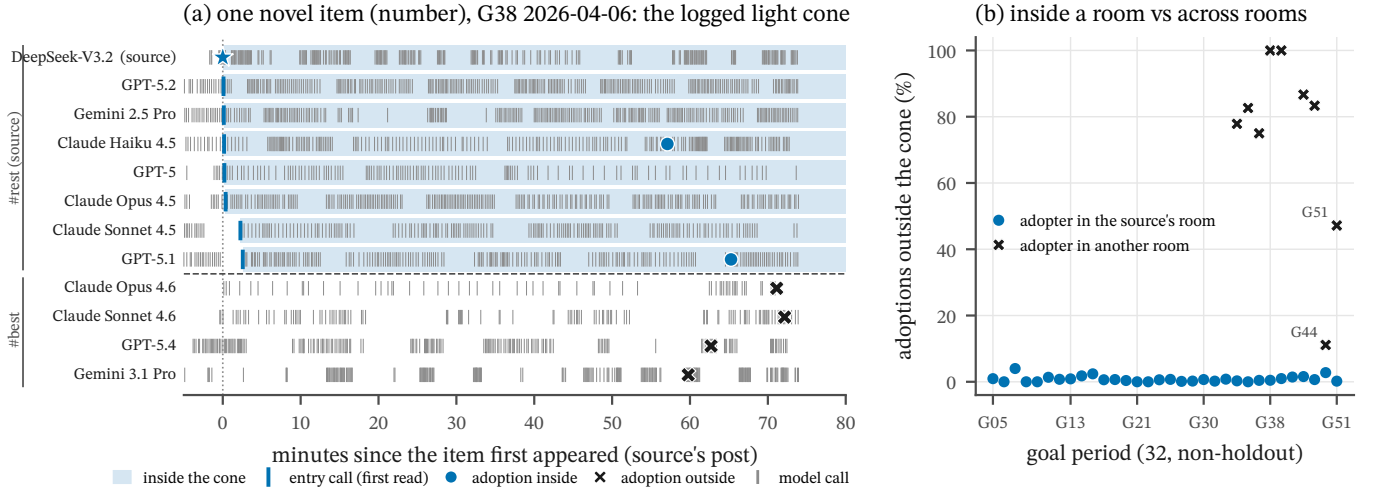


FIG. 4. **Attention dilutes: per-sender uptake falls with the backlog k , and replies go to the newest messages.** (a) The response to a pending sender relative to a no-budget model with the same agent $\times$ day propensities, vs the number k of messages that reached the agent since its previous talk call (context ledger). The plot shows 16 non-holdout periods, with G51 bold and its day-bootstrap 95% band. Every period declines, less steeply than the literal $1/k$ (dotted). (b) The fitted exponent $\hat{\beta}$ (uptake $\propto k^{-\beta}$) on all talk calls (D1, blue) is > 0 in 16/16 periods; the pooled value is 0.66 ± 0.02 (dashed). Gray bands: generative agents with no budget whose replies are triggered (reactive timing, 8 replicates). That null alone gives D1 $\hat{\beta} = 0.75-0.80$, so D1 cannot separate dilution from timing. Timer-wake batches (D2, pink: exogenous k) are not fooled by the null (-0.28 to 0.19). D2 is clear in G51 (0.45 [0.41, 0.50]), G37 and G44, and it contradicts D1 in G38 and G41. (c) G51: the chance that a pending message becomes the reply parent, by its queue rank (1 = newest), for $k = 5-8$ and $9-16$, with day-bootstrap 95% bands. Lines are the rank-blind rates.



SUBCRITICAL SWARMS, FIELDS AND SCALING

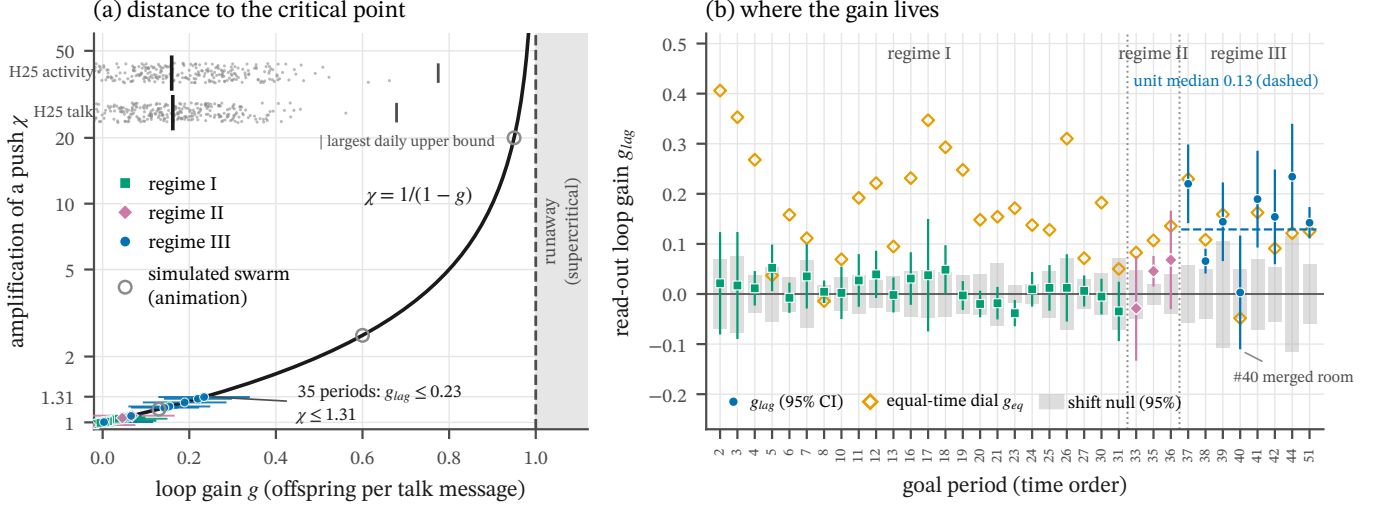


FIG. 6. The talk loop sits far below its critical point. (a) Amplification of a push, $\chi = 1/(1 - g)$, against loop gain g ; every goal period is placed at its read-out loop gain g_{lag} (H67, random-effects pool, 95% CI), and the gray strips are H25's daily equal-time Curie–Weiss dial for activity and talk (282 non-holdout days, medians 0.16; no day's upper bound reaches 0.8). The largest period gain is 0.23, so $\chi \leq 1.31$, against a divergence at $g = 1$; open circles mark the simulated gains of the companion animation. (b) g_{lag} per period in time order: about 0 in regimes I and II, unit median 0.13 in regime III (every unit's upper bound < 0.40), with the shift null (gray, 95%) and the equal-time dial g_{eq} (orange), which reads 0.05–0.40 from fast shared fields where $g_{lag} \approx 0$. Animation: H67-subcritical-dial/anim.mp4.

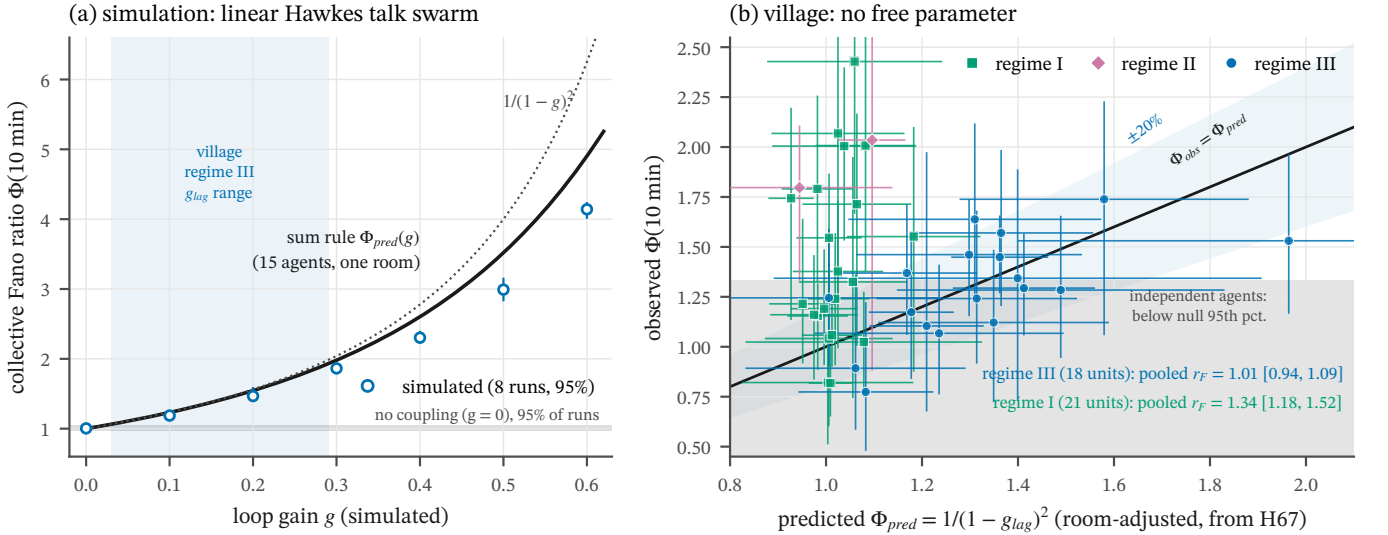


FIG. 7. A fluctuation–response sum rule for talk. (a) Simulation: in a linear Hawkes talk swarm (15 agents, one room, 1-min read-out kernel) the collective Fano ratio of 10-min talk counts, $\Phi = \text{Var}(\sum_i n_i) / \sum_i \text{Var}(n_i)$, follows the room-adjusted prediction $\Phi_{pred}(g) \approx 1/(1 - g)^2$ with no free parameter; the gray band is the no-coupling spread. The estimator reads slightly low at large g because cascades straddle window edges, which is small in the village range ($g_{lag} \leq 0.29$). (b) Village: observed $\Phi(10 \text{ min})$ per unit against Φ_{pred} built from H67's g_{lag} (95% CIs both ways; 41 units with ≥ 40 usable windows). Regime III closes, pooled $r_F = \Phi_{obs}/\Phi_{pred} = 1.01$ [0.94, 1.09] over 18 units; regime I carries an unexplained fast excess, $r_F = 1.34$ [1.18, 1.52]; gray shading marks values below the independent-agent (block-shift) null's 95th percentile.

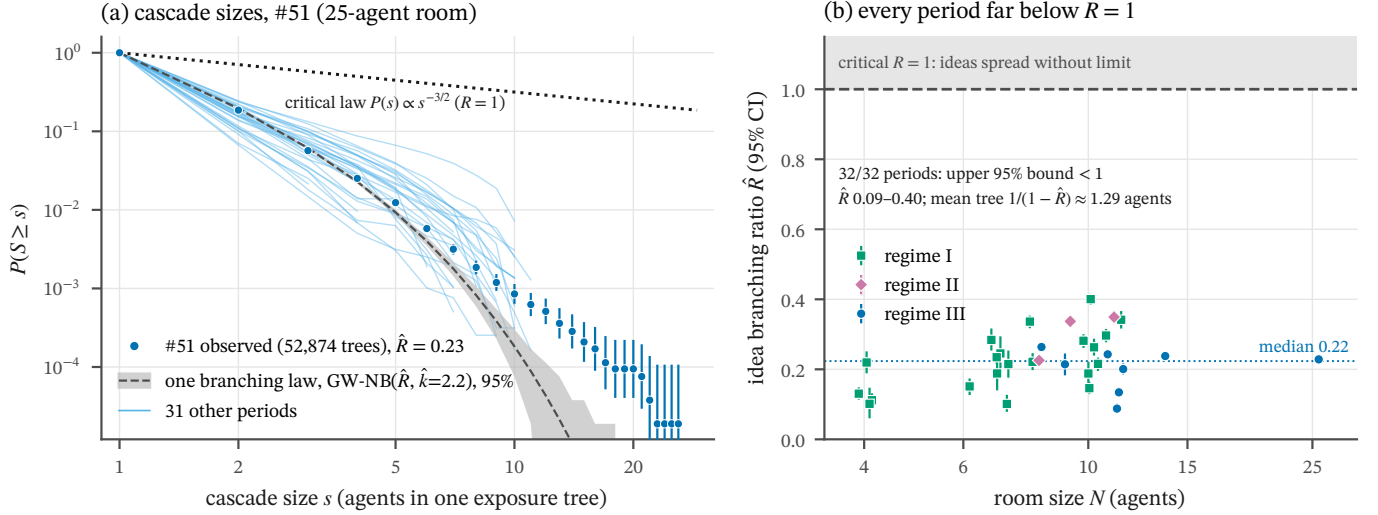


FIG. 8. Idea cascades are strongly subcritical and heavier-tailed than one branching law. (a) Complementary distribution of exposure-tree sizes in #51 (52,874 trees in a 25-agent room; Wilson 95% intervals), against a single Galton-Watson law with negative-binomial offspring at the measured $\hat{R} = 0.23$, $\hat{k} = 2.2$ (gray band: 95% sampling envelope at the observed tree count) and against the critical law $P(s) \propto s^{-3/2}$ ($R = 1$); the other 31 periods are thin lines. Trees of five or more agents are more frequent than the homogeneous law allows, consistent with idea-specific R , but nowhere near critical. (b) Branching ratio $\hat{R}$ (95% CI) for all 32 non-holdout periods against room size: 0.09–0.40, median 0.22, every upper bound below 1. Animation: H34-idea-cascades/anim.mp4.

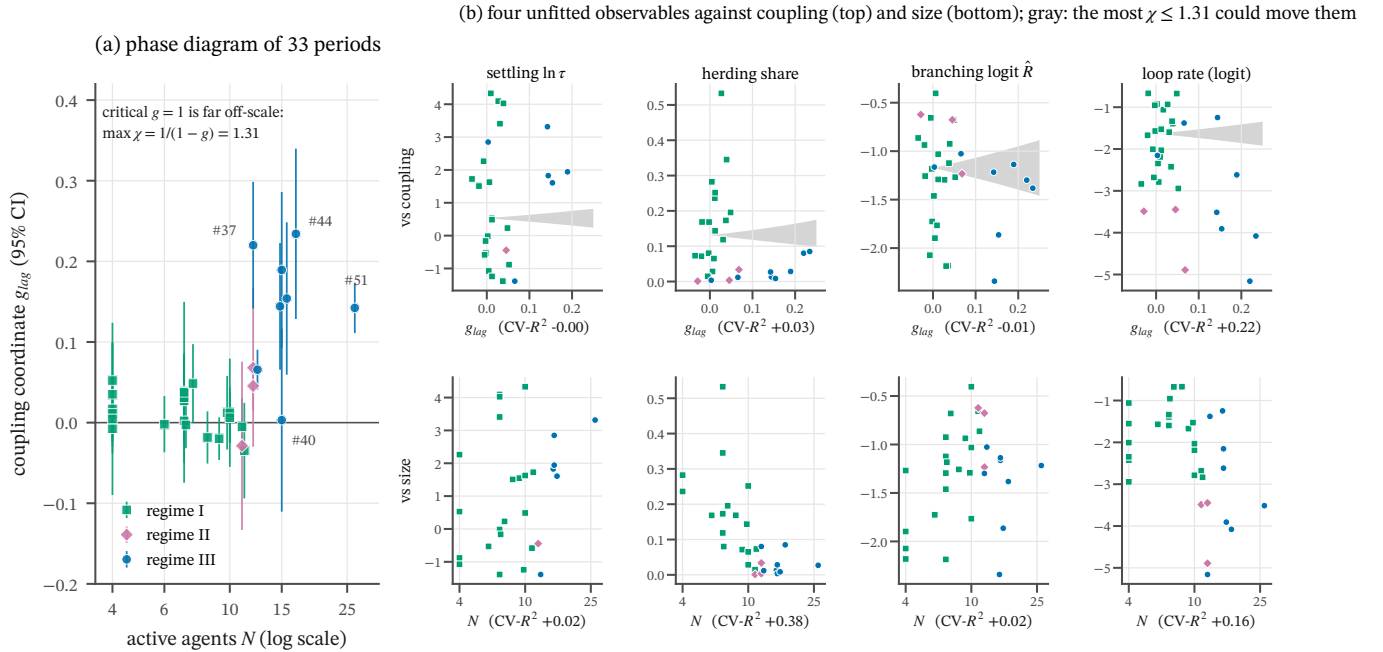


FIG. 9. One dial does not collapse the phase diagram. (a) The 33 non-holdout periods placed by active population N (log scale) and read-out coupling g_{lag} (H67, 95% CI), with scaffold regime as marker; the largest coupling gives $\chi = 1/(1 - g) = 1.31$, so the critical point is far off-scale. (b) Four unfitted observables (settling time, herding share, idea branching, loop rate) against g_{lag} (top) and N (bottom), with leave-one-period-out $CV-R^2$ in each axis label: g_{lag} collapses none of them ($-0.00, 0.03, -0.01, 0.22$; within-regime permutation $p = 0.07$), while $\log N$ carries herding (0.38). The gray band is the most a mean-field amplification $\chi \leq 1.31$ could move each observable from its $g_{lag} \approx 0$ median; the observables vary $\times 3$ to $\times 127$ across periods.

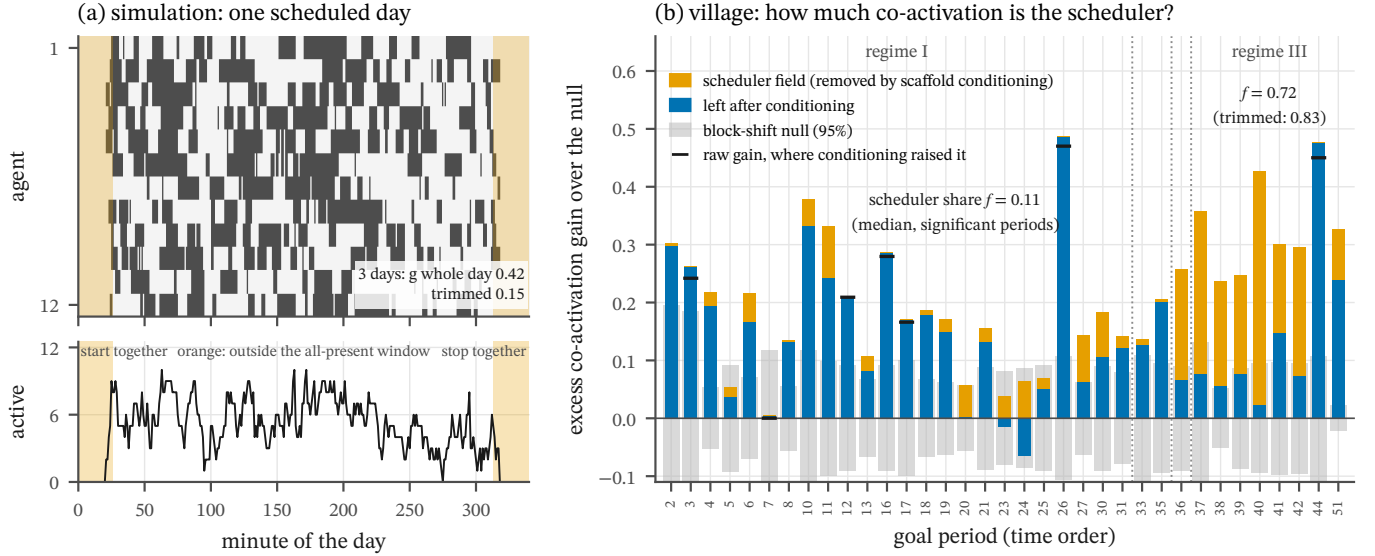


FIG. 10. Scheduler or coupling? (a) Simulation of one scheduled day (12 agents, weak kinetic-Ising coupling): agents start and stop within minutes of the operator's resume and pause (orange: outside the all-present window), and with the 30-min block estimator these shared edges alone inflate the equal-time co-activation gain (three simulated days: 0.42 whole day vs 0.15 inside the window). (b) Village, round 1b: excess activity co-activation gain over the block-shift null (gray, 95%) per period, split into the part removed by scaffold conditioning (orange: off-schedule minutes dropped, not-started, finished and consolidating agent-minutes imputed) and the part that survives (blue). In regime III the median scheduler share is $f = 0.72$ (0.83 after the DQ8 all-present trim; with trim and conditioning together only #44 and #51 keep a significant excess); in regime I it is $f = 0.11$. Unlabelled blocks between the dotted lines are regime II (#33, #35) and the split period #36. Animation: H38-platform-stalls/anim.mp4.

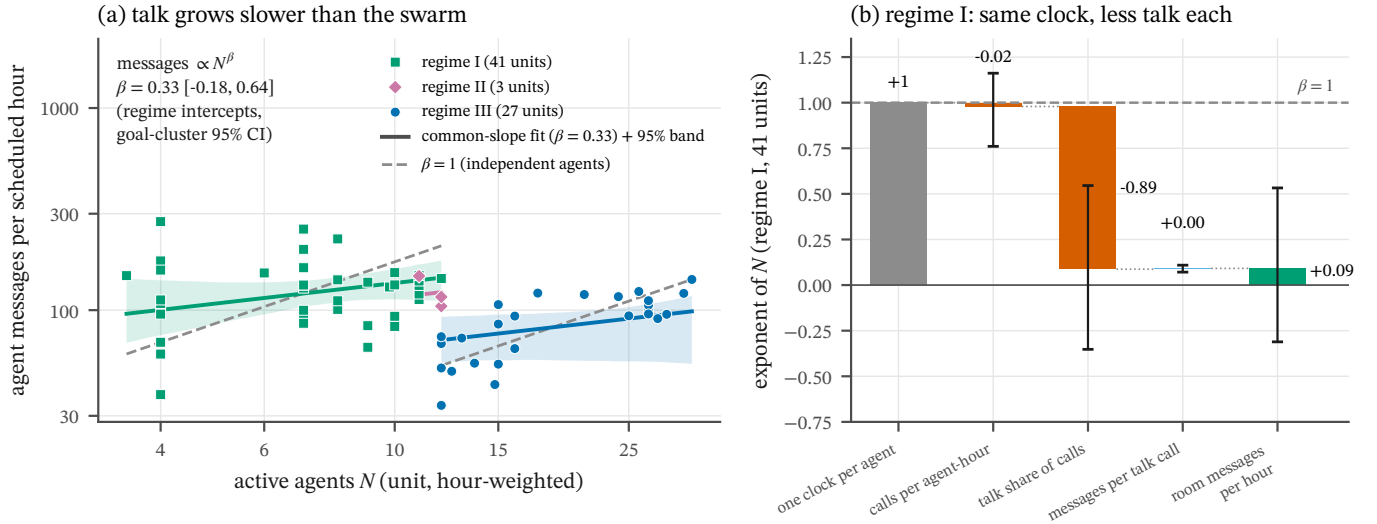


FIG. 11. Talk output scales sublinearly with swarm size. (a) Agent messages per scheduled hour against active population for 71 non-holdout units, with regime intercepts and a common slope: $\beta = 0.33 [-0.18, 0.64]$ (goal-cluster bootstrap), against the independent-agent reference $\beta = 1$ (dashed); within regimes the slope is 0.09 (I) and 0.77 (III). (b) Regime I decomposition of the exponent: each agent adds a clock (+1), calls per agent-hour do not change (-0.02), the talk share of calls falls (-0.89), and messages per talk call are flat, so room talk is nearly fixed (+0.09 [-0.31, 0.53]); error bars are goal-cluster 95% intervals.

GOALS AS FIELDS: QUENCH, REMANENCE,
MEMORY

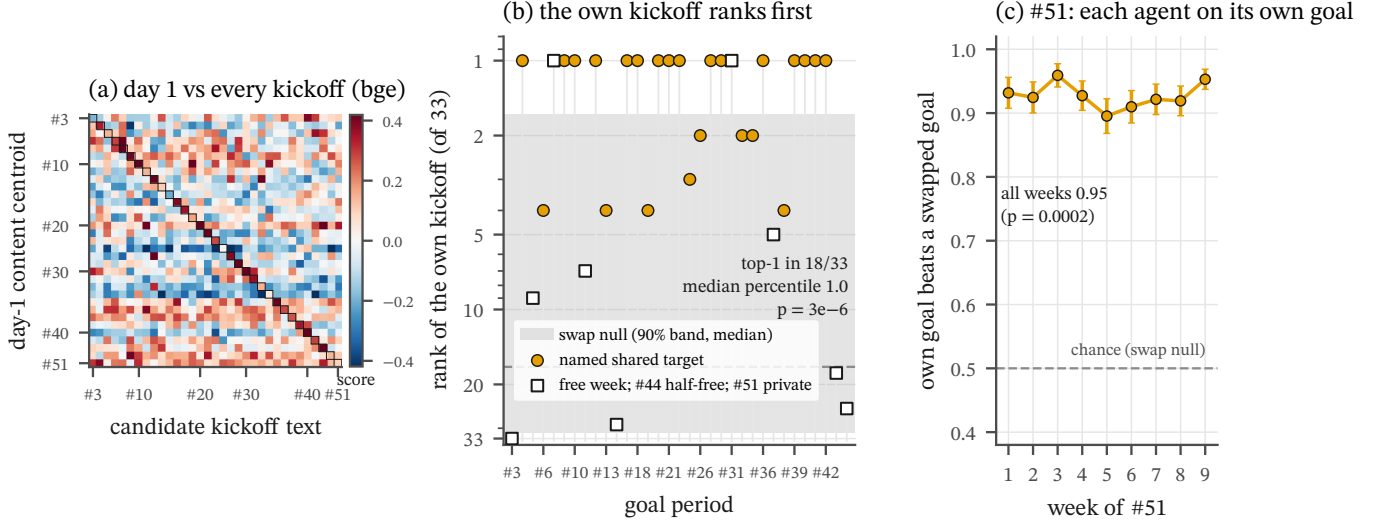


FIG. 12. **Day-1 content lands on its own kickoff text (H54).** (a) Target score of each period's day-1 content centroid (rows) against each of the 33 eligible kickoff texts (columns), bge, regime-whitened, after the genericness correction (a kickoff that resembles every centroid gets no credit); the diagonal is the own kickoff. (b) Rank of the own kickoff among the 33 candidates per period (1 = best). Gray: kickoff-swap null (uniform rank; 90% band and median). The own kickoff ranks first in 18/33 periods (median percentile 1.0, $p = 3 \times 10^{-6}$); the misses are kickoffs that name no shared target (free weeks #3, #16; #44's half-free week; #51's private goals). (c) In #51, where each agent has its own goal text, an agent's content sits closer to its own goal than to a swapped agent's goal in 0.95 of pairs ($p = 0.0002$), flat over nine weeks (weekly 0.90–0.96; bars: binomial 90% CI; dashed: chance).

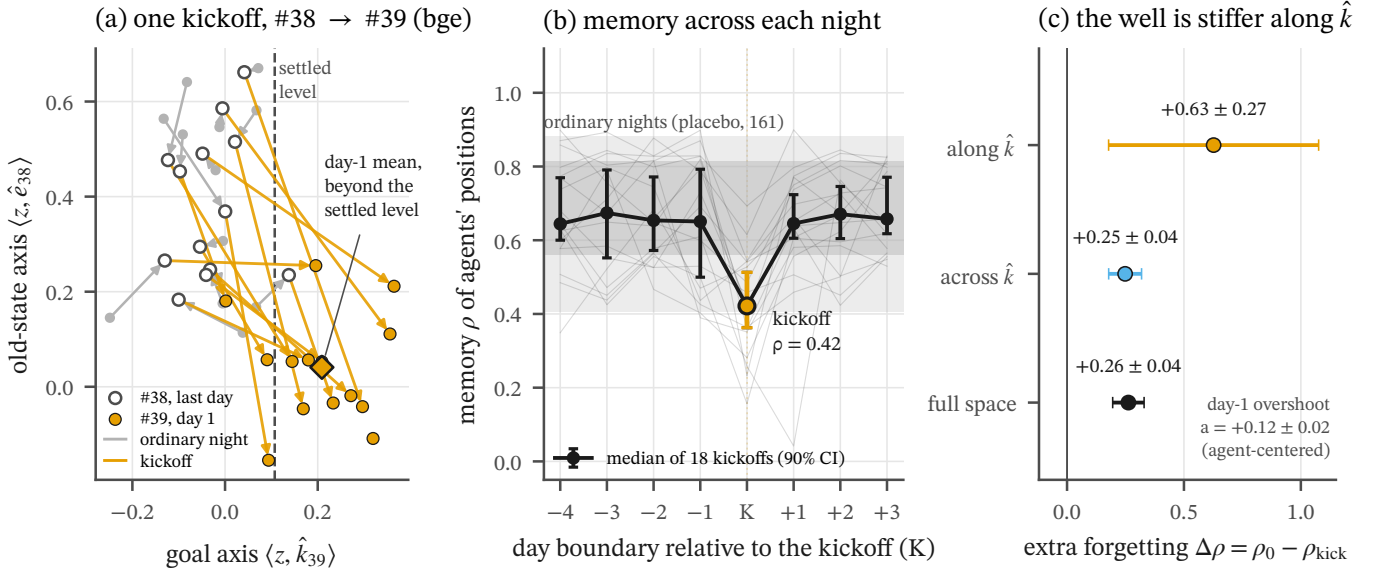


FIG. 13. **A goal kickoff is a restoring force: it erases where each agent was, more along the goal than across it, and overshoots on day 1 (H97).** (a) One real kickoff (#38 → #39, bge statement vectors, regime-whitened): agent day means projected on the new kickoff direction $\hat{k}_{39}$ and on the old-state direction $\hat{e}_{38}$ (the #38 last-day centroid orthogonal to $\hat{k}_{39}$). Gray arrows: an ordinary night inside #38; orange arrows: the kickoff. The day-1 mean (diamond) lies beyond the settled level of days 2–5 (dashed). (b) Cross-agent memory ρ (disattenuated correlation of agent positions before and after a day boundary) around 18 kickoffs: median and bootstrap 90% CI (black; thin lines, single kickoffs); gray, the 161 ordinary-night placebo boundaries (dark: IQR, light: 5–95%). Memory falls from $\rho_0 = 0.68$ to $\rho = 0.42$ at the kickoff only, $\Delta\rho = +0.26 \pm 0.04$, positive in 16/18. (c) Extra forgetting by subspace (random-effects mean, 90% CI): along $\hat{k}$ 0.63 ± 0.27 , across $\hat{k}$ 0.25 ± 0.04 , full space 0.26 ± 0.04 ; the agent-centered day-1 overshoot intercept is $a = +0.12 \pm 0.02$ (0 under the linear law). Animation: H97-quench-restoring-force/anim.mp4.

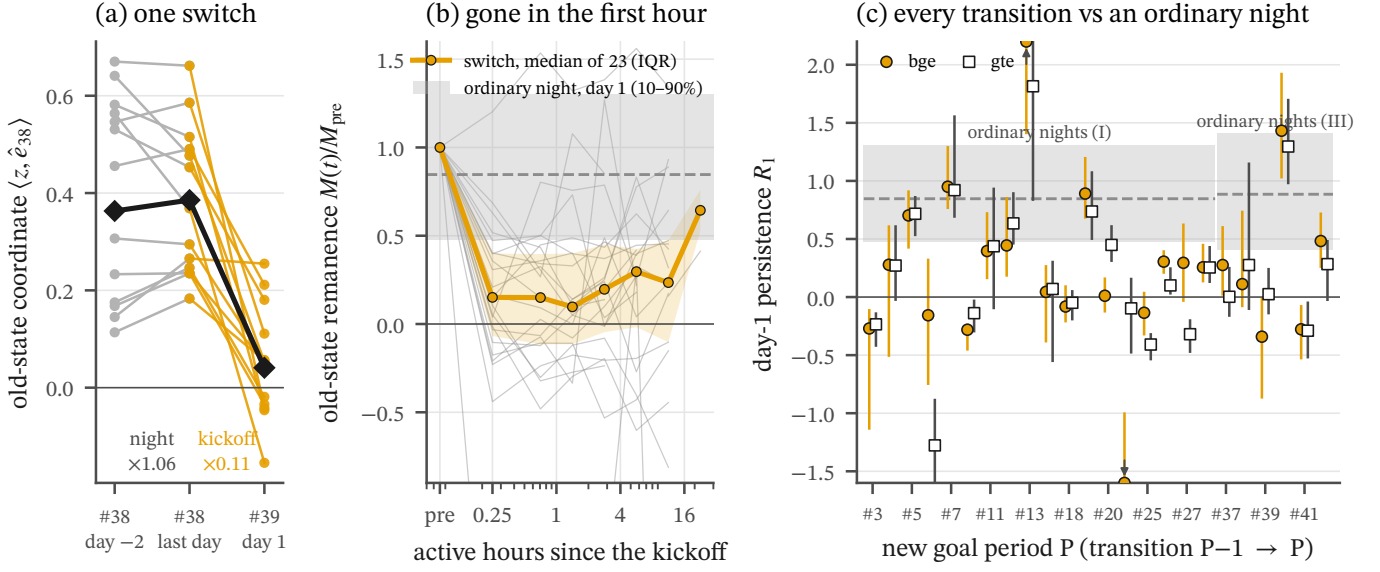


FIG. 14. **A goal switch is a quench: the old state is gone within the first active hour (H96).** (a) Illustration in the shared content plane ($\#38 \rightarrow \#39$, bge): each agent's old-state coordinate $\langle z, \hat{e}_{38} \rangle$ keeps its level across an ordinary night (gray) and collapses across the kickoff (orange); diamonds, swarm means. (b) H96's field-orthogonal old-state remanence $M(t)/M_{\text{pre}}$ against active hours since the kickoff (bge, style-residualized; thin lines, the 23 transitions with $M_{\text{pre}} > 0$; orange, median and IQR). Gray band: day-1 persistence across an ordinary night (146 pseudo-switches, 10-90%; dashed, median 0.85). (c) Day-1 persistence R_1 per transition (bge filled, gte open; 90% CIs, clipped at the axis) against the ordinary-night reference by regime (median 0.85 in regime I, 0.89 in regime III). Median R_1 is 0.27 (bge) / 0.18 (gte); $\Delta R_1 = -0.62 [-0.78, -0.45]$ / $-0.72 [-0.88, -0.56]$, below the night reference in 20/23 / 21/24 transitions.

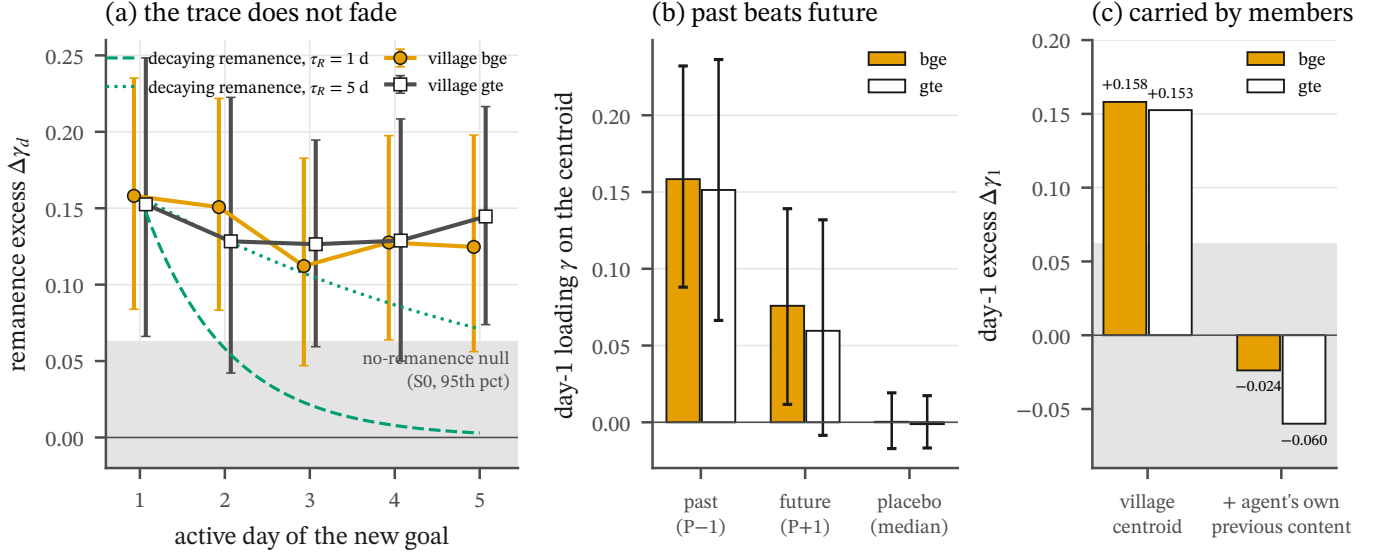


FIG. 15. **The previous goal leaves a day-1 trace that does not decay within five days and is carried by each agent's own past (H82).** (a) Remanence excess $\Delta \gamma_d$ (day- d loading on the previous period's village centroid beyond kickoff, operator, prior and placebo centroids) over 25 goal boundaries, mean and boundary-bootstrap 95% CI, bge (filled) and gte (open). Gray: the synthetic no-remanence null (S0, 95th percentile). Green: the decaying remanence the hypothesis predicted ($\tau_R = 1$ and 5 active days). Day 1 is 0.158 / 0.153, about $3\times$ the null 95th percentile (0.055 / 0.063), and the trace stays near 0.11-0.15 through day 5. (b) Day-1 loadings on the past (P-1), future (P+1) and placebo centroids: 0.16 / 0.15, 0.08 / 0.06 and 0.00; the asymmetry is partly the slow mode's lag geometry. (c) Post hoc: with the agent's own previous-period content as a regressor, the village-centroid excess drops to -0.024 / -0.060 .

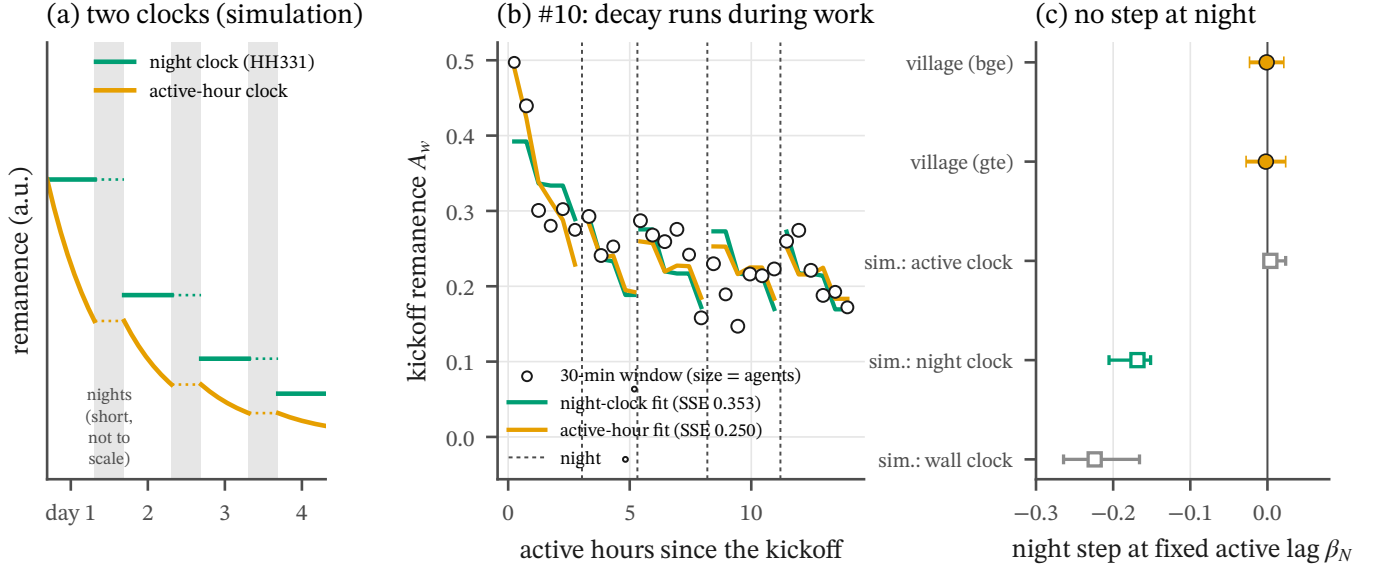


FIG. 16. **Nights do not demagnetize: kickoff remanence decays with active work, not per night (H103).** (a) Simulation of the two candidate clocks: a night clock (one multiplicative step per night, flat during the day) and an active-hour clock (decay during work, flat over the night); nights drawn short. (b) Kickoff remanence A_w in 30-min windows of goal period #10 (bge, style-residualized; symbol size, agents in the window) against active hours since the kickoff, with H103's night-clock (green) and active-hour (orange) fits, both with time-of-day terms and equal parameter counts; dotted lines mark nights. The alignment falls inside each day and partly recovers overnight. Across the 22 periods with decay the active-hour clock fits as well or better in 17/22 (bge) and 16/22 (gte); the night clock wins in 5/22 / 3/22 (synthetic clock-selection accuracy 0.82). (c) Night step at fixed active lag β_N : village random-effects mean over 41 period units, $-0.001 [-0.023, +0.021]$ (bge) and $-0.002 [-0.028, +0.024]$ (gte), against H103's synthetic truths (median, 10–90% of 20 replicates): active-hour clock ≈ 0 , night clock -0.17 , wall clock -0.22 .

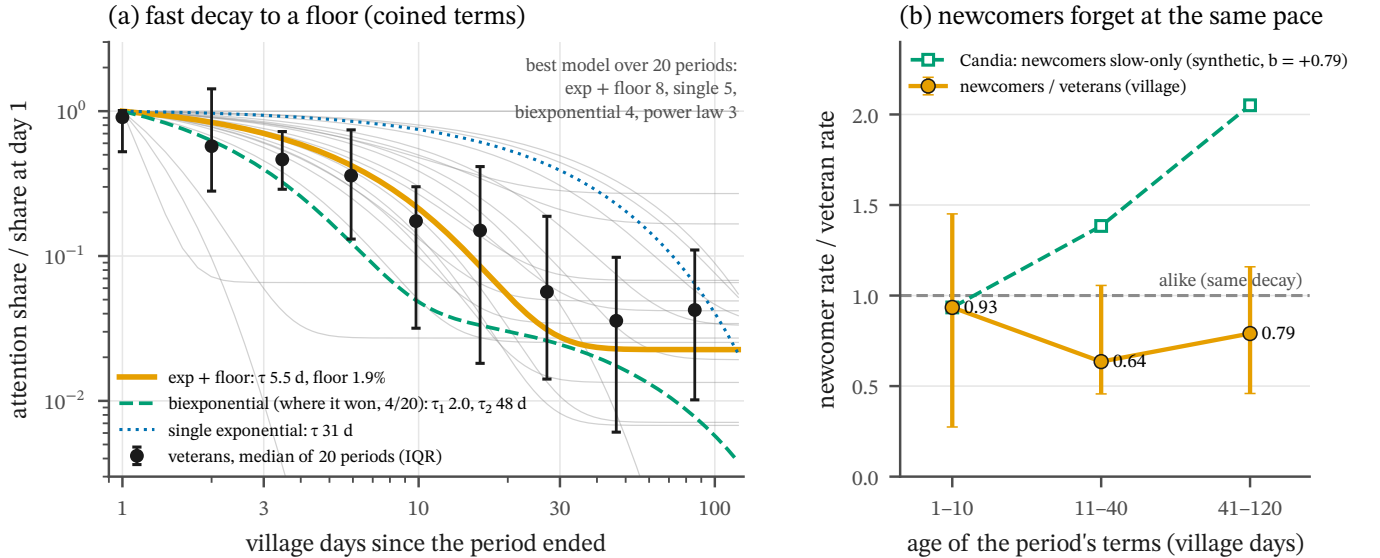


FIG. 17. **Collective memory of a finished goal period decays fast to a small floor, and newcomers forget at the veterans' pace (H88).** (a) Veterans' attention share to a finished period's coined terms against village days since the period ended, each period normalized by its own fitted share at day 1 (thin gray: per-period exponential-plus-floor fits; black: median over 20 periods in day bins, IQR bars). Curves at the median fitted parameters: exponential plus floor ($\tau = 5.5$ village days, floor 1.9% of the initial share; best model in 8 of 20 periods), biexponential where it won ($\tau_1 = 2.0$, $\tau_2 = 48$ d; 4/20) and a single exponential (5/20). (b) Newcomers' standardized attention relative to veterans' at the same age of the terms (indirect standardization, period-bootstrap 95% CI): 0.93, 0.64, 0.79 by age band, with no rise ($b = -0.17 [-0.80, +1.32]$). Dashed gray: equal decay; green: the Candia two-compartment prediction that newcomers carry only the slow component (synthetic, $b = +0.79$; power 0.87–0.92).

ROOMS AS MAGNETIC DOMAINS

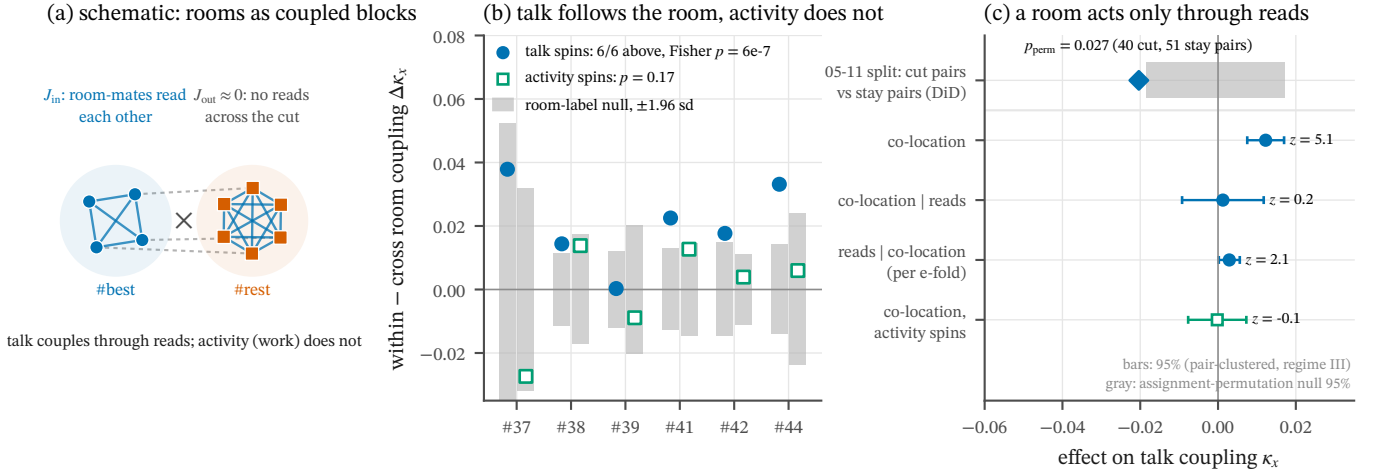


FIG. 18. Rooms decouple chat only through what agents read (H05). (a) Schematic of the two-block kinetic Ising picture: room-mates read each other (J_{in}), and the cut removes the cross-room reads ($J_{\text{out}} \approx 0$). (b) Within-room minus cross-room excess lagged coupling κ_x (over the cross-day surrogate) in the six regime-III two-room windows; gray: room-label null, ± 1.96 sd. Talk coupling follows the room in 6/6 windows (Fisher $p = 6 \times 10^{-7}$); activity spins, the actual work, do not ($p = 0.17$). (c) At the 05-11 split (NE42, not held out) the 40 cut pairs lose talk coupling relative to the 51 pairs that stay together (DiD -0.020 , assignment-permutation $p = 0.027$; gray: permutation null, 95%). Pooled over regime-III pair-days, co-location raises κ_x by 0.012 ($z = 5.1$); with ledger read counts in the model it drops to 0.001 ($z = 0.2$) while reads keep $z = 2.1$, and activity spins show no co-location effect. On the holdout, the 03-16 split (NE15) gave a pair DiD of -0.022 $[-0.038, -0.008]$ ($p = 0.021$), but the pre-registered verdict is inconclusive.

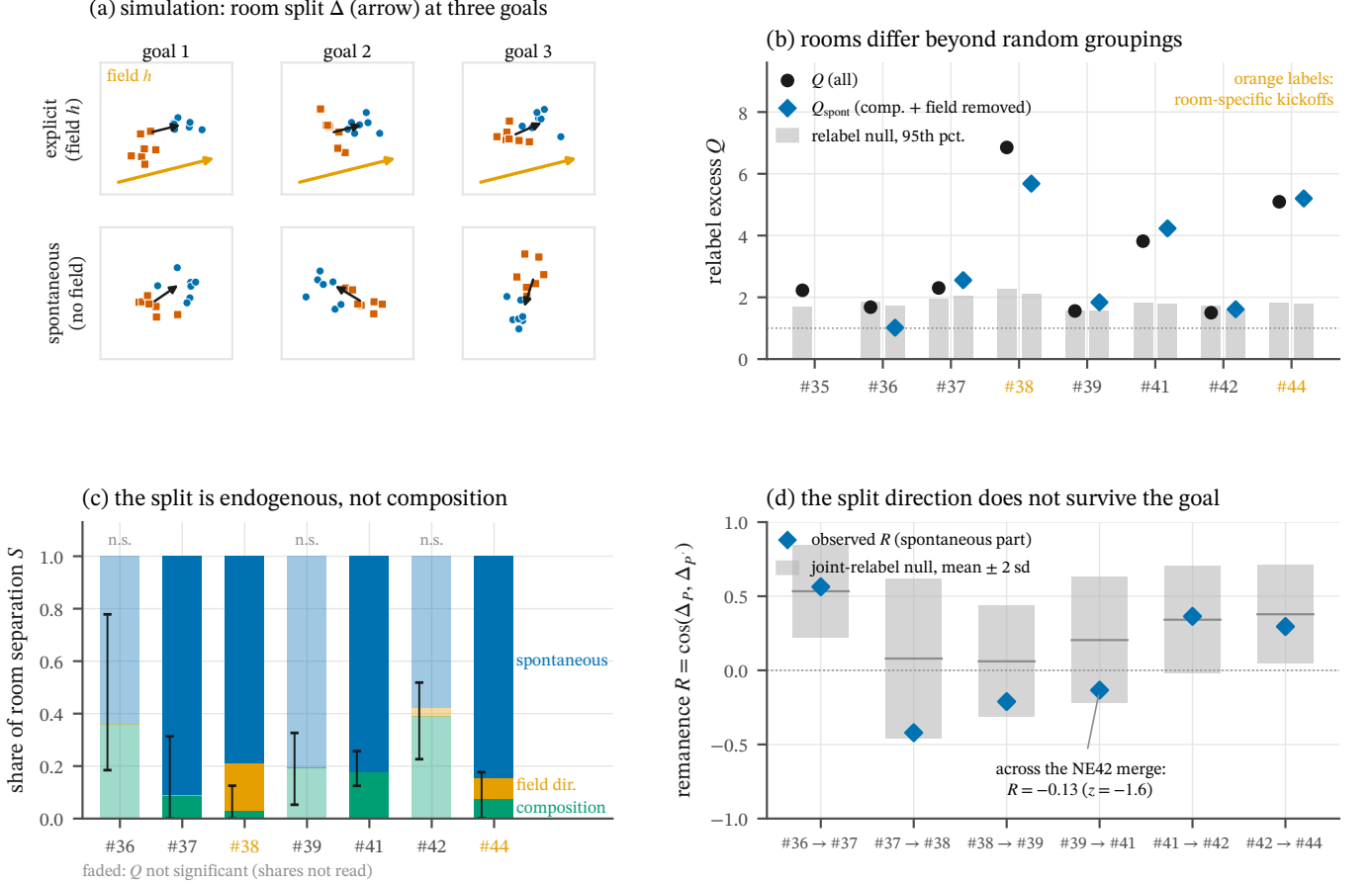


FIG. 19. The $\#_{\text{best}}/\#_{\text{rest}}$ content split is spontaneous and regenerates at each goal (H100). (a) Simulation of a two-block soft-spin model: with a room field h the split Δ points along h at every goal (explicit symmetry breaking); without one it takes a new direction at each goal (spontaneous). (b) Relabel excess Q of the split-half room separation per period, before and after removing composition and the measured field directions (Q_{spont}); gray: room-relabel null, 95th percentile. After both removals the rooms still differ beyond random groupings in 5/7 regime-III periods (Q_{spont} 1.8–5.7). The two periods with room-specific kickoffs (orange) separate most (Q 6.85, 5.09), but their split lies along the kickoff-text direction only at chance (#38: 0.18 vs null 0.16). (c) Shares of the separation: composition 0.03–0.18 where rooms separate significantly, spontaneous 0.58–0.91 (faded: Q not significant; bars: agent-bootstrap 95% CI of the composition share). (d) Remanence R of the spontaneous split direction between consecutive periods against the joint-relabel null (gray: mean ± 2 sd): all six pairs have $|z| < 2$, and across the NE42 merge $R(\#39, \#41) = -0.13$ ($z = -1.6$). bge, style-residualized agent-day vectors. Animation: H100-room-symmetry-breaking/anim.mp4.

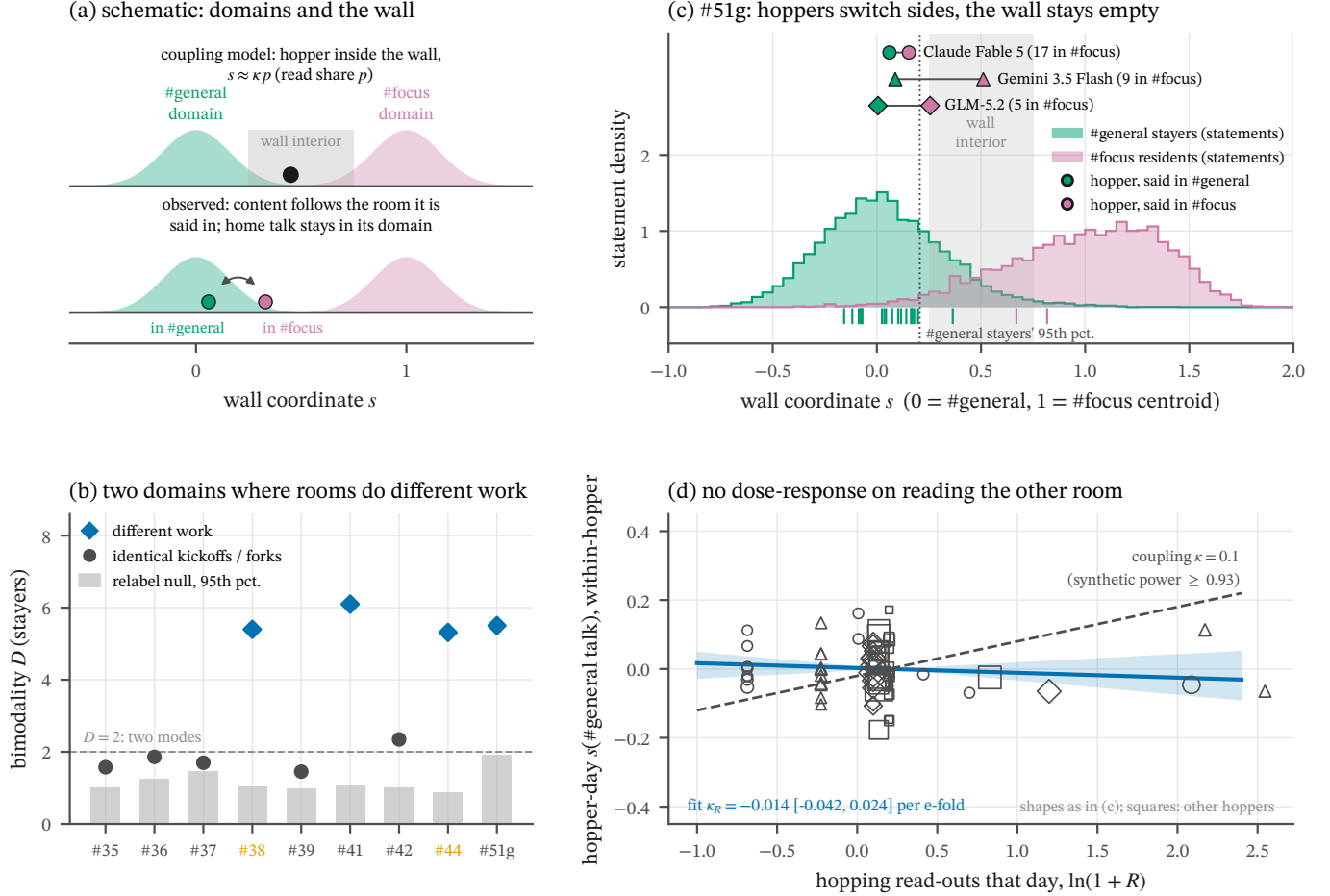


FIG. 20. Two content domains without a wall: a hopper's talk follows the room it is said in (H102). (a) Schematic: the coupling model puts a hopper inside the wall ($s \approx \kappa p$, with p its read share from the other room); observed, its home-room talk stays in its own domain. (b) Bimodality D of the stayers' positions on the inter-room axis per unit; gray: room-relabel null (axis refitted), 95th percentile. Rooms with different work form two domains (D 5.3–6.1 in #38, #41, #44 and 5.5 in #51g; relabel $p \leq 0.006$); identical-kickoff weeks split less (median 1.70). (c) Unit #51g: wall coordinate s of the #general stayers' and #focus residents' statements, the stayers' leave-one-out positions (rug) and the hoppers' mean positions for statements made in #general (green) and in #focus (pink). The one frequent hopper, Claude Fable 5, sits at $s = 0.06$ in #general talk, inside the stayers' range (95th percentile 0.21); hoppers' statements made inside #focus sit at s 0.15–0.51. (d) Hopper-day s of #general talk against hopping read-outs (within-hopper): $\kappa_R = -0.014 [-0.042, 0.024]$ per e-fold (band: day-bootstrap 95%), where the synthetic power against $\kappa = 0.1$ (dashed) is ≥ 0.93 . bge, style-residualized statement vectors. Animation: H102-room-domain-walls/anim.mp4.

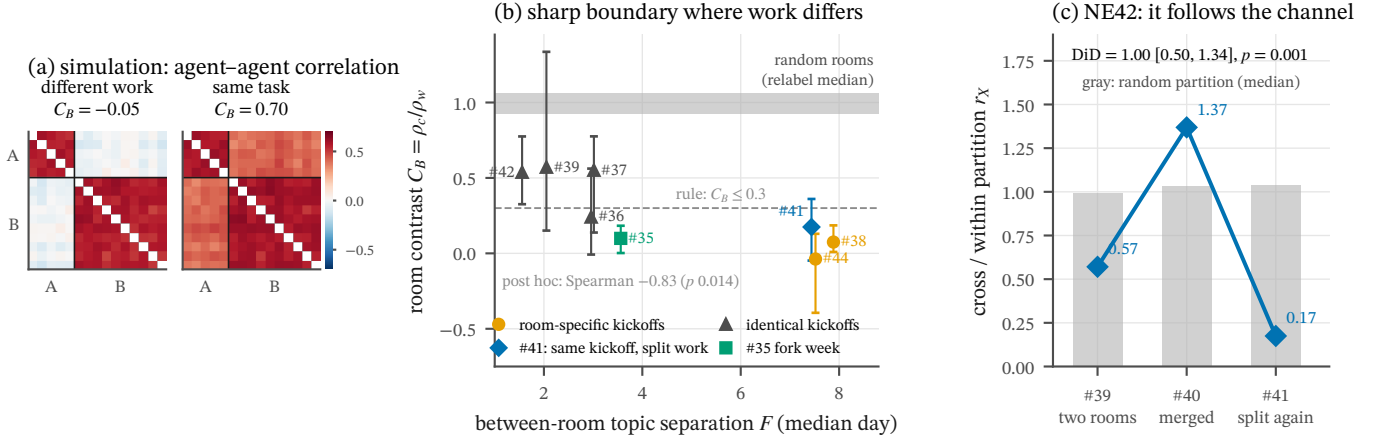


FIG. 21. Rooms bound content coherence when they do different work, and the boundary follows the channel (H47). (a) Simulation of agent-agent content correlations in a 5 + 10 two-room world: room-specific drives give a block matrix ($C_B \approx 0$), one shared task drive does not. (b) Room contrast $C_B = \rho_c/\rho_w$ (cross-room over within-room per-pair correlation, 30-min windows; bars: day-bootstrap 95% CI) against the between-room topic separation F ; gray: room-relabel null medians; dashed: the $C_B \leq 0.3$ rule, met with $p < 0.05$ in 6/9 periods (median 0.23). The sharpest boundaries are the room-specific-kickoff weeks (#38 0.07, #44 -0.04) and #41, where identical kickoffs led to split work (0.17); identical-kickoff weeks share about half their fluctuation (median 0.54). The separation axis is post hoc (Spearman -0.83 , $p = 0.014$). (c) NE42 merge and split: cross-partition over within-partition correlation r_X is 0.57 with two rooms, 1.37 when merged and 0.17 after the split (DiD 1.00 [0.50, 1.34], $p = 0.001$); gray: partition-permutation null medians. bge only.

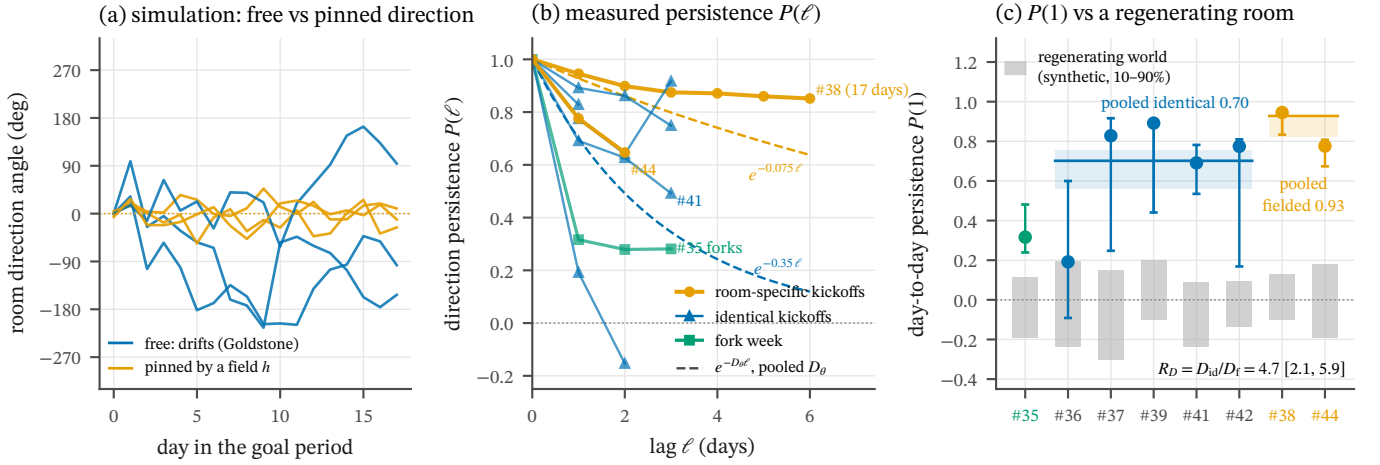


FIG. 22. The room content direction persists within a goal period, more so with room-specific kickoffs (H108). (a) Simulation of the room order-parameter angle over 17 days: free Goldstone diffusion at the identical-kickoff rate ($D_\theta = 0.35$ per day) vs an angle pinned by a field ($D_\theta = 0.075$ per day). (b) Measured noise-corrected direction persistence $P(\ell)$ (joint-relabel excess) per period, lags with at least two day pairs; dashed: $e^{-D_\theta \ell}$ for the pooled groups. #38 (17 days) falls from 0.95 to a plateau of 0.85 at six days, the pinned signature. (c) Day-to-day persistence $P(1)$ with agent-bootstrap 95% CIs (which sit low) against the regenerating synthetic world (gray, 10–90%): $P(1)$ is 0.69–0.95 in 6/8 periods; the fork week (#35, 0.32) and #36 lose the direction. Pooled, identical-kickoff periods keep 0.70 [0.56, 0.75] and room-kickoff periods 0.93 [0.82, 0.92]: $R_D = 4.7$ [2.1, 5.9] (gte 5.2). The contrast rests on #38 (without it $R_D = 1.4$) and is confounded with split size. bge, style-residualized.

THE CONTEXT WINDOW CARRIES THE
VALUE

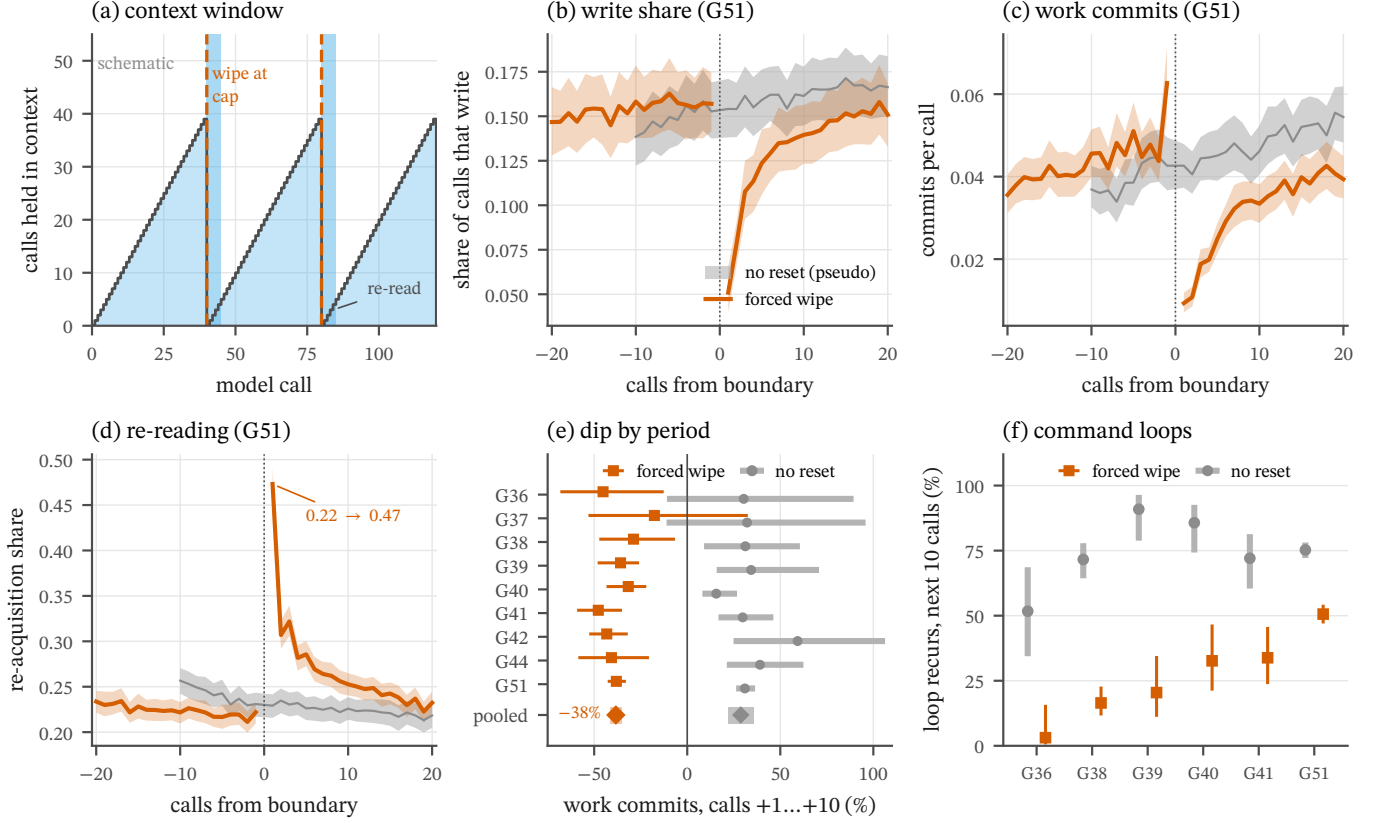


FIG. 23. A forced context wipe costs output and starts a short re-reading phase. (a) Schematic: the scaffold wipes the context window at the 41-record cap (40 calls); the agent then re-reads. (b)–(d) Measured in G51 (14,593 forced wipes), calls aligned to the wipe (vermillion, 95% agent-day bootstrap band) and to a no-reset boundary at call 21 of a segment (gray band, the null). At the first call after a wipe the re-acquisition share rises from 0.22 to 0.47 and relaxes over about 5–7 calls; writes and work commits dip and recover over about 3 calls. (e) Work commits over calls +1...+10 relative to calls -20...-11, per period: pooled -38% [-41, -35] after forced wipes (H15's ledger estimator: -39% [-42, -35], 7/9 periods), against a no-reset control that rises because output ramps up through every segment. (f) A command loop in the last 10 calls recurs in the next 10 calls far less often after a wipe (G38: 16% vs 72%; pooled OR 0.11 [0.05, 0.26], 7 periods; bars are binomial 95% intervals). Animation: H44-context-wipe/anim.mp4.

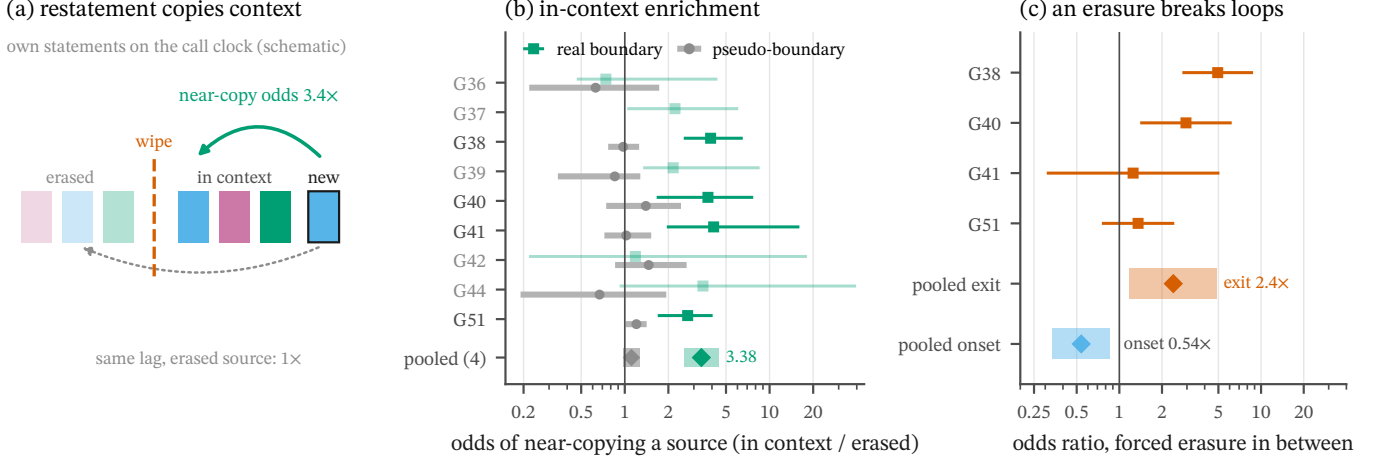


FIG. 24. Restatement loops copy what is still in the context window. (a) Schematic: an agent’s own statements on the call clock; a wipe removes the older ones from its context. (b) Mantel-Haenszel odds that a statement near-copies an earlier own statement in the current context segment rather than one an erasure removed, at matched agent, lag and call distance (green; agent-day bootstrap 95% intervals; faded periods have fewer than 30 loop episodes). The gray null puts a fake boundary at each segment’s midpoint. Pooled over the four scorable periods the enrichment is 3.38 [2.56, 4.46] ($I^2 = 0$), against 1.11 [0.98, 1.26] at the pseudo-boundary. (c) A forced erasure between two statements raises the odds that a loop ends 2.40 $\times$ [1.18, 4.88] and lowers the odds that one starts to 0.54 $\times$ [0.34, 0.86] (logits with agent effects, agent-day cluster errors; random-effects pools). Animation: H69-restatement-loops/anim.mp4.

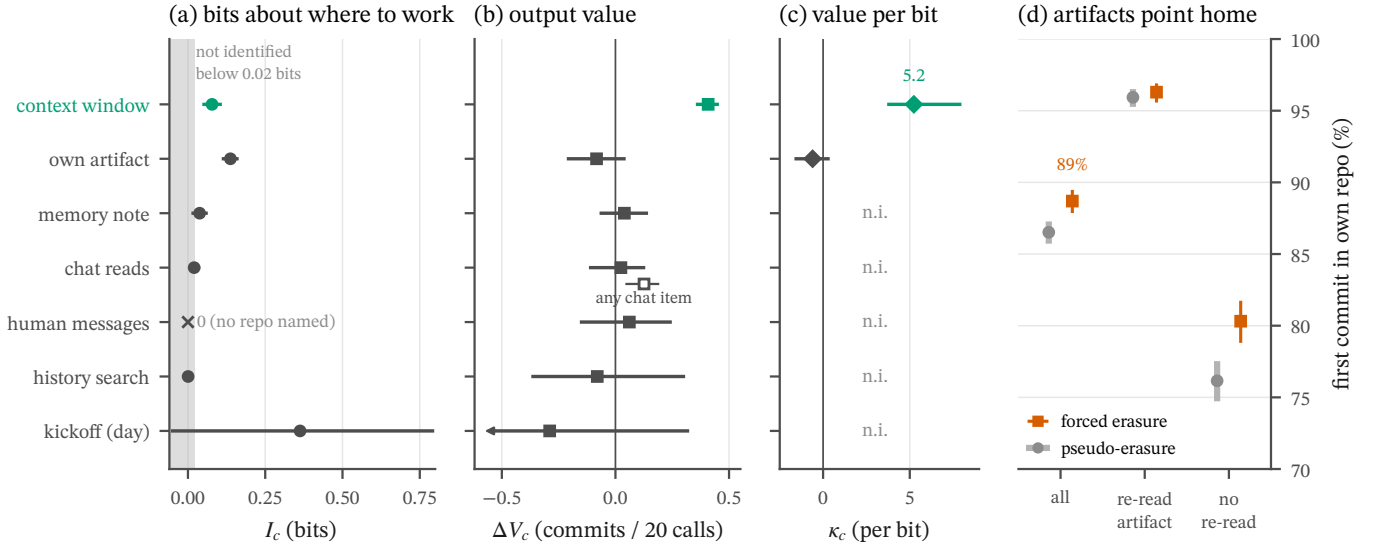


FIG. 25. The κ table: only the context window buys commits per bit. Each row is a channel scored at its natural scramble (NE41 forced erasures, 18,760 events vs 19,886 pseudo-erasures; the kickoff at new-goal nights, NE34). (a) Allocation information I_c (bits about the repo of the next commit, beyond agent identity; the within-agent permutation floor is subtracted). Rows whose lower bound is under 0.02 bits (gray) are not identified. (b) Value ΔV_c in work commits per 20 calls: erasing the context costs 0.41 [0.36, 0.45] (41% of output); every other row’s interval includes 0. Receiving any chat item after an erasure (hollow) adds output without repo bits. (c) $\kappa_c = \Delta V_c / I_c$ where both rows are identified: $\kappa_C = 5.2$ [3.8, 7.9], $\kappa_A = -0.6$ [-1.6, +0.3], $P(\kappa_C > \kappa_A) = 1.00$; “n.i.” = not identified. (d) The artifact tells the agent where to work: 89% of first commits after a forced erasure land in its own repo (87% after a pseudo-erasure), and re-reading the artifact raises that to 96% in both arms, so it adds no output beyond its placebo association. Bars: agent-day paired bootstrap 95% intervals (a–c), binomial intervals (d).

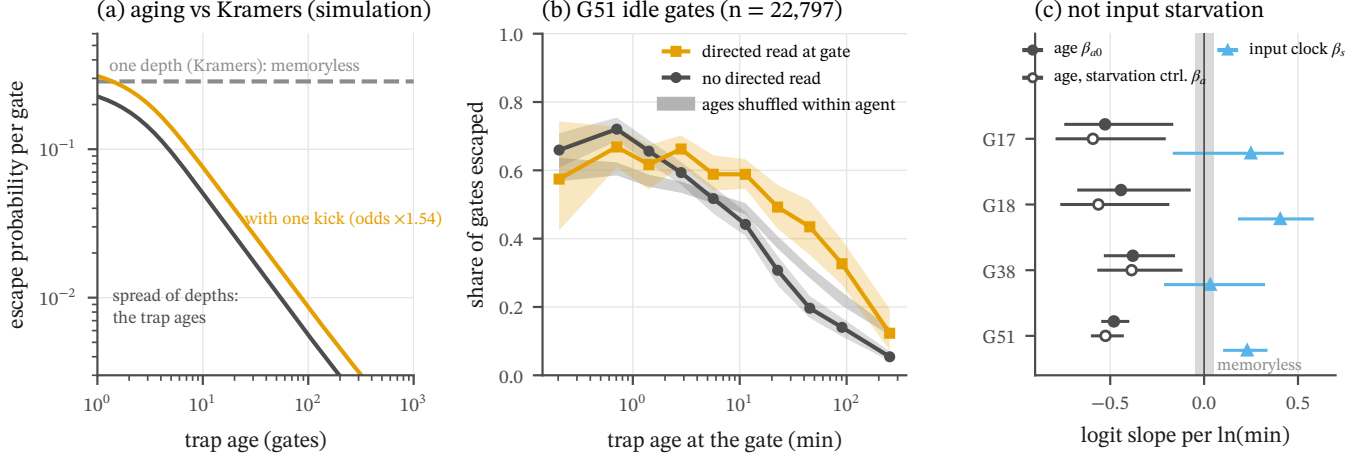


FIG. 26. Idle traps age by themselves, and one directed kick helps. (a) Simulation: with one trap depth (Kramers) the escape probability per gate does not depend on the time already spent in the trap; with a broad spread of depths it falls with trap age. A kick that multiplies the escape odds by 1.54 shifts the aging curve up without removing the aging. (b) G51 idle gates (22,797; a gate is a call after an idle call): share that start a sustained run, by trap age, for gates with a directed read (mention or own nudge, orange) and without (dark), with agent-day bootstrap 95% bands. The gray band is the null with trap ages shuffled within each agent, which keeps each agent's escape rate and age mix; the steeper fall of the data is within-agent aging. (c) Fixed-effect logit slopes on $\ln(\text{trap age})$ in the four periods with powered aging, without (β_{a0}) and with (β_a) the input-starvation clock, and the input clock itself (β_s): in G51 $\beta_{a0} = -0.48 [-0.54, -0.41]$, $\beta_a = -0.53 [-0.59, -0.44]$, $\beta_s = +0.23 [+0.11, +0.33]$, so aging is not input starvation (gray band: memoryless). A directed kick at the gate raises the odds of acting $1.5\text{--}2.9\times$ (G51 1.54; H16). Animation: H16-trap-aging/anim.mp4.

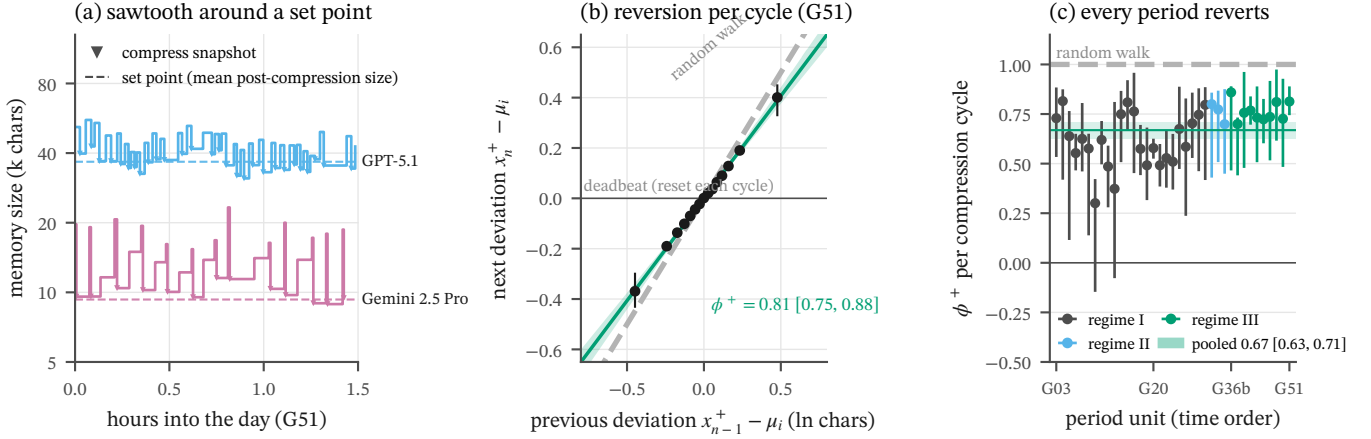


FIG. 27. Memory size is held by a homeostat on the consolidation clock. (a) Memory size of two G51 agents over 90 minutes: append snapshots grow the memory file and each consolidation compresses it (triangles) back toward the agent's set point (dashed: its mean post-compression size in G51). (b) G51 post-compression deviation from the agent set point against the previous cycle's deviation (26,681 cycles, 15 quantile bins, agent-bootstrap 95% intervals). The fitted persistence $\phi^+ = 0.81 [0.75, 0.88]$ (green) lies between a random walk (slope 1, gray null) and a deadbeat reset (slope 0); the next cycle does not overshoot to the other side. (c) ϕ^+ in all 37 non-holdout period units: the interval excludes 1 in 37/37, pooled 0.67 [0.63, 0.71]. Forced and voluntary consolidations give the same ϕ^+ ($|F - V| < 0.15$ in 9/10 regime-III periods). The loop is not first order (AR(2) median +0.19): a fast compression tracks a set point that drifts slowly (persistence ≈ 0.82 per cycle).

**COLLECTIVE ORDER: HERDING,
ALLOCATION, CULTURE**

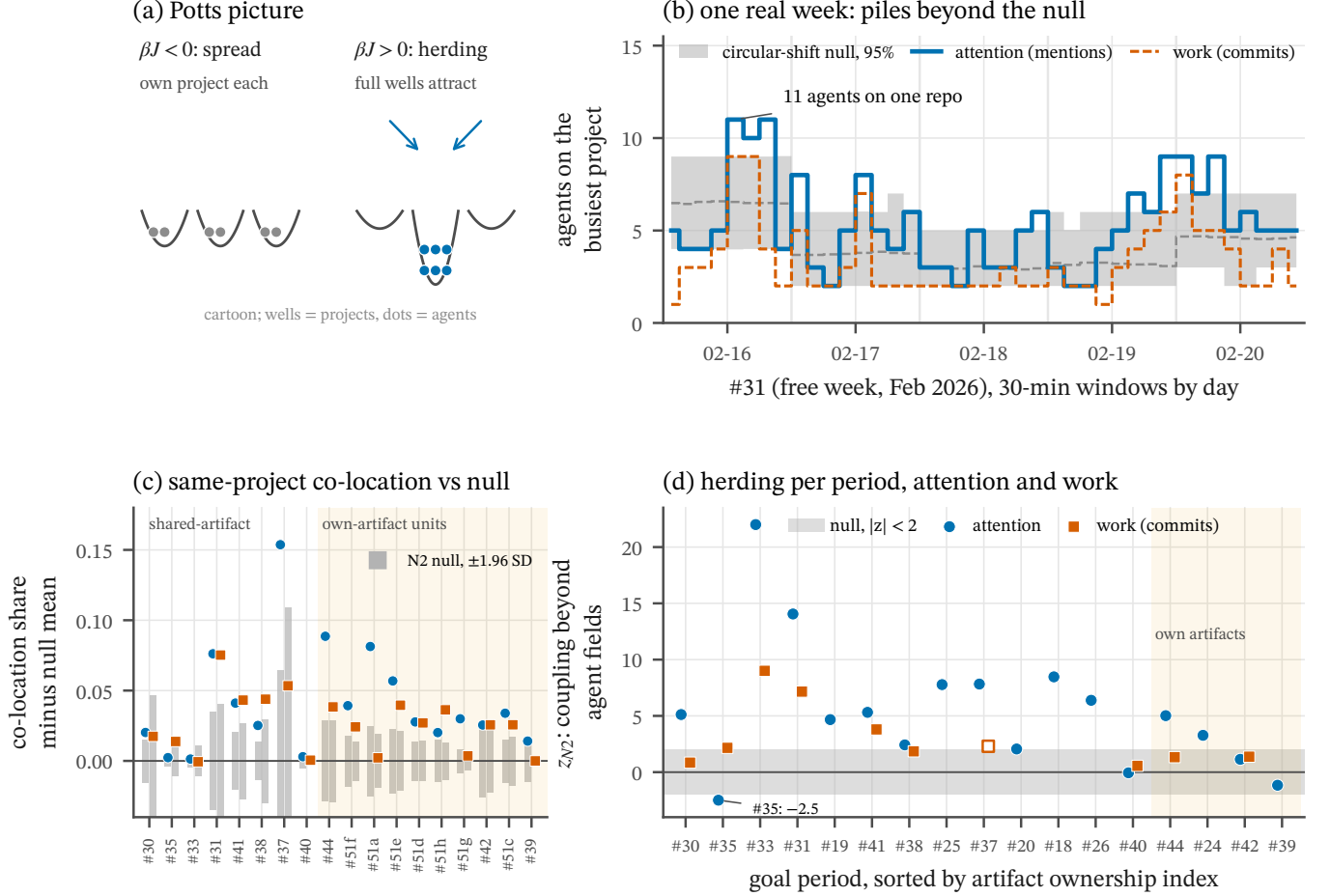


FIG. 28. **Agents herd onto shared projects, in attention and in work commits (H11).** (a) Curie-Weiss Potts cartoon: projects are wells and agents are balls; $\beta J < 0$ spreads agents over projects, $\beta J > 0$ piles them into one. (b) Goal period #31 (free week): agents on the busiest project per 30-min window, from strict artifact mentions (blue) and from agent work commits (red, dashed); gray band, 95% interval of the circular-shift null N_2 , which keeps each agent's own persistence; up to 11 agents mention one repo in one window and 9 commit to one. (c) Same-project co-location share minus its N_2 mean, per unit from #30 on (gray bars, null ± 1.96 SD); the excess is about equal in attention and work (#31: +0.07 in both). (d) Coupling beyond agent fields, z_{N_2} , per period, sorted by artifact ownership; gray band, $|z| < 2$. Attention herds ($\beta J_{CW} > 0$) in 11 of 14 tested weeks whatever the goal mode, with $z_{N_2} \geq 2$ in 10 of 11 shared-artifact weeks; work herds ($z_{N_2} \geq 2$) in 4 of 6 testable shared weeks (open square: #37, too few work blocks). The exceptions are own-artifact weeks (spread by agent fields, not avoidance) and #35 ($z_{N_2} = -2.5$, a shared week against the pattern). Animation: [H11-herding-shared-projects/anim.mp4](#).

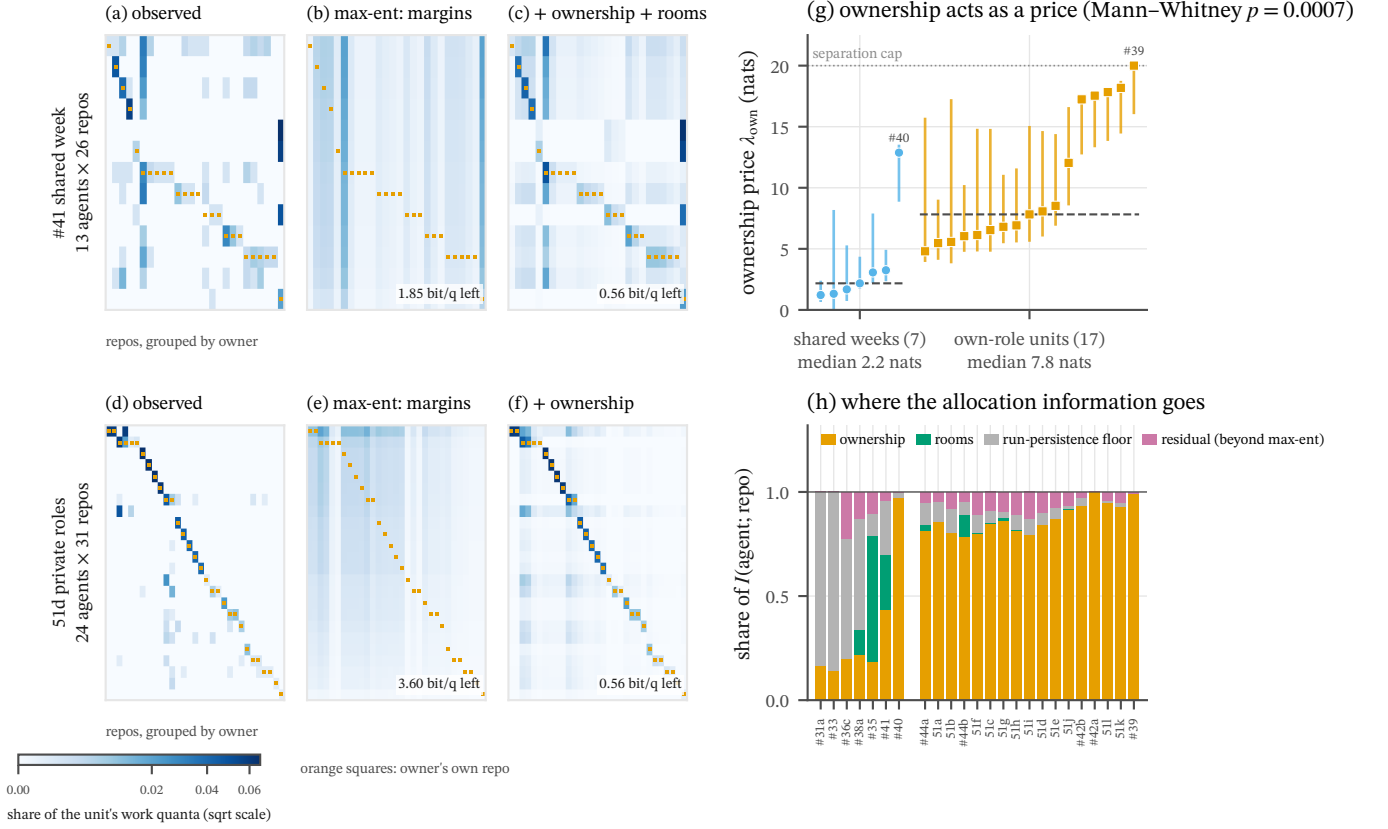


FIG. 29. **Work is allocated at maximum entropy once activity, repo sizes, ownership and rooms are fixed (H94).** (a–c) Goal period #41 (shared week): observed agent $\times$ repo work quanta (30-min agent-windows with ≥ 1 work commit), the max-ent table with agent and repo margins only (M1, $D_1 = 1.85$ bits per quantum left), and the max-ent table that also matches ownership and room totals (M3, 0.56 bits left). (d–f) The same for private-role unit 51d with ownership (M2: 3.60 $\rightarrow$ 0.56 bits). Orange marks are each repo's owner. (g) Ownership price λ_{own} per testable unit with agent-bootstrap 95% intervals: median 2.2 nats in shared weeks and 7.8 in own-role units (Mann-Whitney $p = 0.0007$); #40, a hub beside own worlds, sits at 12.9. (h) Share of $I(\text{agent}; \text{repo})$ carried by ownership, by rooms, by the run-persistence floor (gray, the null: a max-ent draw that keeps each agent-day's runs) and left as residual beyond it (pink): ≤ 0.13 in 23 of 24 units (#36c 0.23).

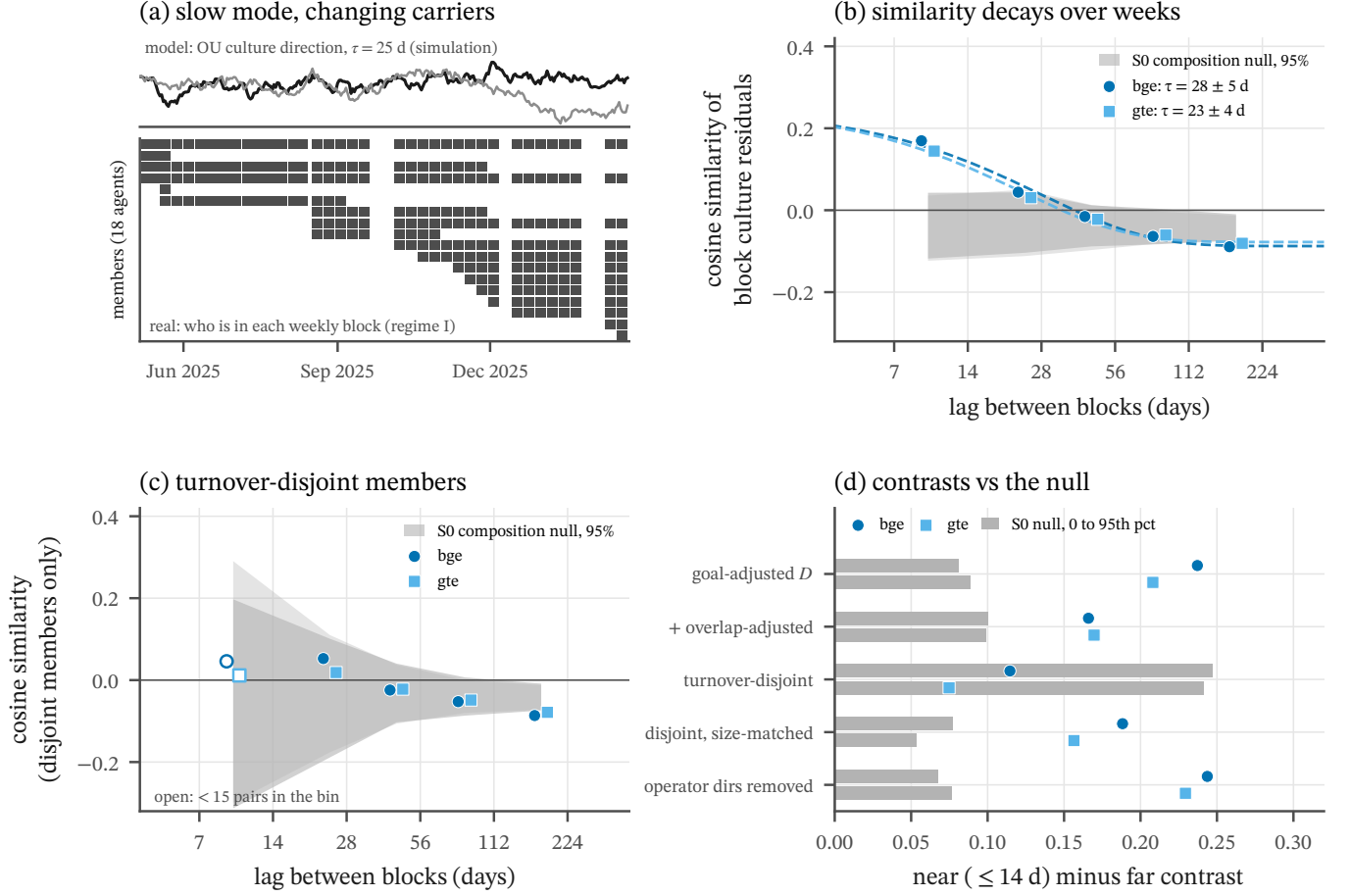


FIG. 30. **Regime-I village content carries a slow collective mode beyond its members and the measured fields (H81).** (a) Top: the model, an Ornstein-Uhlenbeck culture direction with $\tau = 25$ d (simulation). Bottom: who is present in each weekly block of regime I (May 2025–Feb 2026, real); the carriers turn over almost completely. (b) Cosine similarity of block culture residuals (members' leave-goal-out personal vectors and the goal, kickoff and human-message directions removed) for block pairs in different goal periods, against lag, for two embedding models; gray, 95% band of the S0 composition null on the real panel (personal vectors plus exchangeable goal and block fields, no memory). Blocks ≤ 14 d apart align ($\bar{s} = 0.17 / 0.14$) and the similarity decays with $\tau_u = 28 \pm 5 / 23 \pm 4$ d (dashed, OU fit). (c) The same using only members absent from the other block (open symbols: < 15 pairs); this raw turnover-disjoint statistic has little power (S0 band wide). (d) Near-minus-far contrasts against the S0 95th percentile (gray bars, bge upper, gte lower): goal-adjusted 0.237 / 0.208 vs 0.081 / 0.089; overlap-adjusted 0.166 / 0.170 vs 0.100 / 0.099; turnover-disjoint 0.115 / 0.075 (fails, power ≤ 0.4) but 0.188 / 0.156 when size-matched ($m = 4$; vs 0.078 / 0.054); 0.244 / 0.229 after removing operator directions beyond the centroid (round 2; vs 0.067 / 0.076). The mode is not carried by the long-lived artifact record its members read; an unmeasured calendar-time outside drift is narrowed, not excluded.

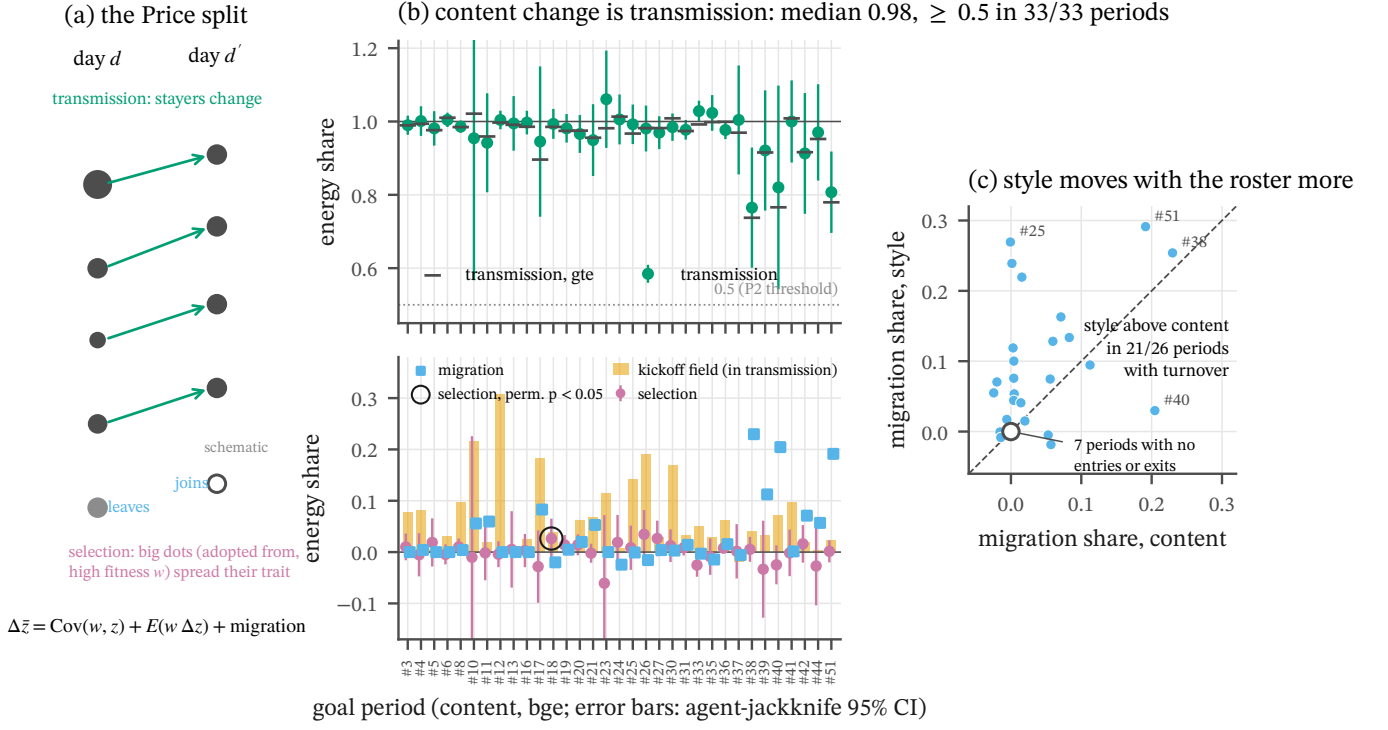


FIG. 31. **Village content changes by transmission; selection is not detectable (H89).** (a) The cultural Price split of the day-to-day change of the mean trait (schematic): stayers change (transmission), stayers whose ideas others adopt inside the logged light cone have higher fitness w (selection), and leavers and entrants move the mean (migration). (b) Cross-fitted energy share of transmission in the change of mean content (bge statement vectors) for the 33 eligible periods, with agent-jackknife 95% intervals; dashes, the second embedding model (gte). Median 0.98 in both models; ≥ 0.5 and the largest term in 33 of 33 periods. Lower (b): the small terms, selection (median $|s_{\text{Sel}}| = 0.010$; beyond its fitness-permutation null in 1 of 33 periods, circled, against 1.7 expected by chance), migration (median 0.004; visible only in the largest, most open periods #38, #40, #51), and the part of transmission along the kickoff direction (orange, the exogenous field; median 0.04). The synthetic validation shows that selection in a vector trait cannot be detected at village counts, so its small share is a bound set by the design, not a discovery. (c) Migration share of style (20 text features) against content: style is above content in 21 of 26 periods with any turnover (open circle: 7 periods with no entries or exits).

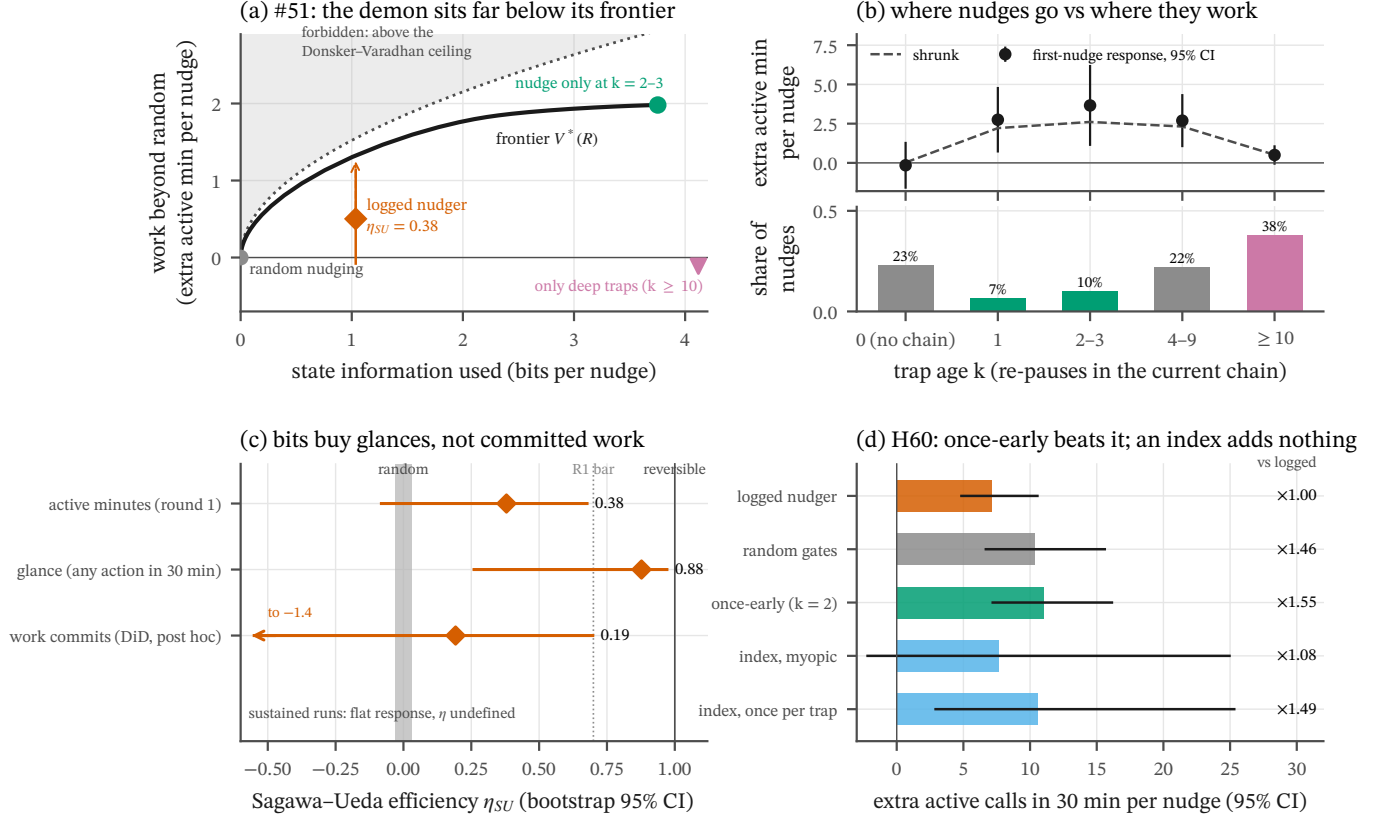


FIG. 32. **The auto-nudger is an inefficient Maxwell demon: its bits buy glances, not committed work (H35, with H60).** (a) Goal period #51 (short-pause scaffold), trap-age state space: work bought beyond a random nudger with the same budget against the state information the policy uses. The logged nudger uses 1.04 bits per nudge and buys $+0.50 [-0.09, 1.16]$ extra active minutes, $\eta_{SU} = 0.38 [-0.08, 0.68]$ of what the frontier $V^*(R)$ gives at the same bits; nudging once at the early re-pauses ($k = 2-3$) reaches the frontier's end ($\times 2.3 [1.7, 5.2]$ work per nudge), nudging only deep traps is worse than random; the shaded region lies above the Donsker-Varadhan ceiling. (b) First-nudge response by trap age (95% CI; dashed, shrunk) and where the nudges go: 38% fall on chains ≥ 10 pauses deep, where a nudge buys 0.5 minutes against 2.2-2.6 at $k = 1-9$. (c) Efficiency by outcome (round 1b, gray band: random nudging): glances $\eta_{SU} = 0.88 [0.26, 0.97]$; committed work 0.19 $[-1.39, 0.70]$ after placebo differencing (post hoc), with $+0.22 [-0.29, 0.82]$ commits per hour per nudge. (d) H60, idle gates in #51: extra active calls in 30 min per nudge under each policy at the logged budget (cross-fitted, 95% CI). Random gate choice beats the logged nudger ($\times 1.46 [1.22, 1.67]$) and once-early beats it ($\times 1.55 [1.25, 1.81]$); a fitted index does not beat once-early ($\times 1.08 [-0.30, 3.44]$; once per trap $\times 1.49 [0.46, 3.42]$).